

Constraint-guided multi-agent deep reinforcement learning for multi-microgrid energy management with mobile electric vehicles

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HIGHLIGHTS

- A microgrid–EV mobility–energy coordination model is formulated for MMG scheduling.
- EV mobility, physical connection, battery degradation, and range-anxiety risk are explicitly modeled.
- A hybrid and constraint-guided MA-DRL framework is developed for online decision-making.
- Total operating cost, carbon emissions, power-balance violation, and EV reliability are effectively improved.

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ABSTRACT

Multi-microgrid systems (MMGs) provide a promising architecture for coordinating distributed renewable generation, energy storage systems, and electric vehicles (EVs) in modern distribution networks, with potential benefits for low-carbon operation and economic efficiency. However, existing MMG energy-management studies often simplify EVs as aggregated controllable loads or stationary storage units, thereby neglecting the coupled effects of mobility-dependent availability, travel behavior, range-anxiety risk, and operational constraints. To bridge this gap, this paper proposes a Hybrid and EV-Aware Constraint-Guided Multi-Agent Deep Reinforcement Learning (HEC-MADRL) framework for coordinated microgrid–EV mobility–energy management. A two-layer optimization model is proposed to couple microgrid-side scheduling with EV-side mobility–energy coordination through aggregated EV charging/discharging power. The coupled problem is then reformulated as a cooperative multi-agent Markov decision process, in which each microgrid acts as an autonomous agent and EV mobility states are updated by the dynamic environment. Building on this formulation, HEC-MADRL is developed under a centralized-training and decentralized-execution paradigm. An EV-aware attention mechanism is introduced to represent time-varying EV populations, while hybrid stochastic policy optimization and a twin-critic learning scheme are designed to handle mixed discrete-continuous dispatch decisions and stabilize cooperative value estimation. Case studies on a residential MMG system using real-world profiles demonstrate that HEC-MADRL outperforms representative model-based and learning-based methods in economic performance, carbon emissions, power-balance feasibility, and EV mobility reliability. Compared with the best-performing baseline, MAAC-PER, the proposed method reduces the total objective value by 4.43%, while requiring only 0.043 s for online decision-making. These results indicate that HEC-MADRL provides a scalable, low-carbon, and mobility-aware solution for online MMG–EV coordination.

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1. Introduction

1.1. Motivation

The increasing penetration of renewable energy resources, distributed energy storage systems (ESSs), and electric vehicles (EVs) is fundamentally reshaping the operational paradigm of modern distribution networks [1–3]. In this context, microgrids have emerged as a flexible and modular platform for coordinating heterogeneous distributed energy resources, including photovoltaic (PV) units, wind turbines (WTs), dispatchable distributed generators (DGs), ESSs, demand response (DR) resources, and utility-grid transactions [4]. Beyond the operation of an isolated microgrid, multi-microgrid systems (MMGs) enable wider spatial coordination among distributed resources, thereby improving renewable-energy accommodation, operational resilience, and local energy balancing [5]. Nevertheless, the scheduling of such systems is becoming increasingly intricate, as renewable generation, electricity prices, storage dynamics, and flexible demand are temporally coupled and subject to multi-source uncertainties [6,7].

The rapid deployment of EVs further enriches the flexibility of microgrid operation, while simultaneously introducing mobility-driven uncertainties that are fundamentally different from those of stationary resources [8,9]. Through coordinated charging and vehicle-to-grid (V2G) services, EVs can function as mobile energy storage resources to mitigate peak demand, support local power balance, and improve renewable-energy utilization [10,11]. However, unlike stationary ESSs, the dispatchability of EVs is constrained by travel behavior, queuing status, and battery state of charge (SOC) [12]. Moreover, aggressive V2G utilization may improve short-term microgrid economy, but it can also accelerate battery energy depletion, increase range-anxiety risk, and compromise user mobility reliability. Therefore, treating EVs as stationary storage devices or aggregated controllable loads may obscure their spatio-temporal availability and service-quality requirements, leading to schedules that are economically attractive but unacceptable from the user side.

Coordinated MMG scheduling with mobile EV participation therefore gives rise to a highly coupled and mobility-dependent decision-making problem. On the microgrid side, heterogeneous resources must be scheduled under economic, low-carbon, and feasibility-oriented objectives. On the EV side, charging/discharging flexibility is not continuously available, but is governed by travel behavior, queuing status, SOC evolution, and departure-energy requirements. This makes the problem fundamentally different from conventional microgrid scheduling with stationary ESSs or aggregated flexible loads. In addition, practical MMG operation involves hybrid decisions, including binary operating states and continuous dispatching commands, while a fully centralized optimization scheme may become computationally expensive and communication-intensive as the number of microgrids, EVs, and operational constraints increases. These characteristics reveal a clear knowledge gap: existing energy-management methods have not yet provided a unified framework that simultaneously captures MMG-side low-carbon operation, EV-side mobility reliability, and hybrid dispatch decisions. This gap motivates the development of a mobility-aware, constraint-guided, and multi-agent learning framework for coordinated MMG–EV energy management.

1.2. Literature review

Existing research on microgrid energy management can be organized along two complementary dimensions. From a methodological perspective, current approaches are broadly divided into model-based optimization and learning-based optimization paradigms. From an application perspective, they differ in whether they address a single microgrid or coordinated MMGs, whether EVs are modeled as aggregated loads or mobile resources, and whether V2G operation is considered together with travel, queuing, connection, and departure requirements. This

two-dimensional taxonomy clarifies both the solution mechanisms and the physical scope of prior studies.

Model-based approaches, represented by robust optimization [13] and model predictive control (MPC) [14], have been widely used to coordinate DGs, ESSs, renewable generation, demand response, and grid transactions under explicit operational constraints. Within this category, some studies focus on single-microgrid scheduling without EV participation, whereas others incorporate coordinated charging or V2G operation by treating EVs as aggregated controllable loads or stationary storage resources with predetermined availability. For example, Xiao et al. [15] developed a data-driven distributionally robust optimization framework for microgrid scheduling that accounts for the strategic behaviors of heterogeneous wind power producers. Tan et al. [16] proposed an MPC-based energy-management strategy for a shipboard direct-current microgrid to suppress bus-voltage fluctuations. Although these methods provide interpretable decisions and explicit constraint handling, they mainly address a single microgrid or simplified EV participation. When coordinated MMG operation, binary device states, uncertainty scenarios, V2G decisions, and mobility-dependent EV availability are considered simultaneously, the resulting optimization problem becomes substantially more complex. In addition, rolling-horizon implementation requires repeated online optimization, which may limit its applicability to fast MMG scheduling.

Learning-based methods have emerged as an alternative because they can learn sequential control policies through environment interactions and provide rapid online decisions after offline training [17,18]. Existing DRL studies can be further classified into battery and renewable-energy scheduling, EV charging and V2G control, mobility-aware EV coordination, and multi-energy microgrid dispatch. Representative applications include battery scheduling [19], EV charging control [20], renewable-energy management [21], and low-carbon multi-energy dispatch [22]. In many EV-related studies, bidirectional charging is included, but EVs remain represented as aggregated resources with fixed or exogenously prescribed connection intervals. Such formulations capture V2G capability but do not fully describe the underlying mobility process. Xiong et al. [23], for example, developed a DRL-based real-time energy-management model for a single microgrid integrating EVs and renewable generation. Although the method supports online EV scheduling, it does not address cooperative coordination among multiple microgrids or explicitly model transitions among traveling, queuing, and connected states. Zhang et al. [24] proposed a physical-perception federated DRL method for multi-energy microgrids with heterogeneous load levels, emphasizing privacy preservation and distributed learning, whereas EV mobility behavior and departure-SOC reliability were not explicitly considered. Wu et al. [5] developed an online reinforcement-learning-based energy-management strategy under weather- and climate-induced uncertainties, but hybrid discrete-continuous decisions and mobility-dependent EV participation remained outside its scope.

Multi-agent DRL extends learning-based energy management to distributed and cooperative settings. Existing MA-DRL studies can be broadly divided into three application categories: device- or charging-station-level coordination within a single system, coordination among multiple microgrids or energy hubs with aggregated EV flexibility, and joint MMG–EV scheduling with mobile EV participation. By assigning autonomous agents to different microgrids, energy hubs, or charging stations, MA-DRL can reduce the dimensionality burden of centralized control while learning cooperative strategies from system-level interactions [25,26]. Recent studies have explored MA-DRL for interconnected microgrids and multi-energy communities [27,28]. Zhou et al. [29] investigated joint energy and carbon trading in MMG–EV systems and developed an enhanced MAPPO-based method with parameter sharing and recurrent feature encoding. This work advances cooperative energy trading and V2G participation, but it does not explicitly incorporate user-side mobility reliability, particularly range-anxiety risk associated with V2G discharging. Xu et al. [30] proposed a

hierarchical MA-DRL framework for distributed integrated energy management in multi-energy MMGs, coordinating inter-microgrid planning and intra-microgrid control under economic and carbon-emission objectives. However, its emphasis is on energy-hub-level interaction, and the time-varying association between mobile EVs and individual microgrids remains insufficiently characterized. Zhang et al. [31] developed a projection-based safe MA-DRL method for decentralized energy management in MMG-EV communities. Although the safety projection improves action feasibility, the EV mobility process, including traveling, queuing, physical connection, and departure-dependent reliability, is not fully integrated into the scheduling formulation.

The above review reveals that a unified MMG-EV mobility-energy scheduling framework is still lacking. Existing studies have not sufficiently explained how EV traveling, queuing, physical connection, departure-SOC requirements, and range-anxiety risk interact with MMG power balance and low-carbon operation. Moreover, the time-varying association between mobile EVs and individual microgrids requires an adaptive EV representation mechanism, rather than a fixed-dimensional input structure. From the decision-making perspective, practical MMG scheduling involves hybrid actions composed of binary operating states and continuous dispatching powers, which cannot be effectively handled by policies designed solely for discrete or continuous control. In addition, high-dimensional inter-microgrid coupling and stochastic EV participation may destabilize cooperative value estimation. These gaps motivate the development of a customized MA-DRL framework that combines EV-aware state representation, hybrid policy learning, and centralized cooperative value estimation for scalable online MMG-EV mobility-energy management.

1.3. Contributions

To address the above challenges, this paper proposes a Hybrid and EV-Aware Constraint-Guided Multi-Agent Deep Reinforcement Learning (HEC-MADRL) framework for coordinated microgrid-EV mobility-energy management. The main contributions are summarized as follows.

- (1) A two-layer microgrid-EV coordination model is formulated to jointly characterize MMG operation and EV mobility reliability. By linking microgrid power balance, low-carbon operation, with EV charging/discharging behavior, the model establishes a unified representation of stationary resources and mobile flexibility for coordinated and efficient MMG scheduling in modern energy systems.
- (2) The coupled scheduling problem is reformulated as a cooperative multi-agent Markov decision process (MDP), in which each microgrid acts as an autonomous agent and EV mobility states are updated by the environment. An EV-aware observation structure is designed by integrating local microgrid states, compact inter-microgrid coordination signals, and attention-based EV aggregation, enabling decentralized agents to represent time-varying EV populations.
- (3) A customized HEC-MADRL algorithm is proposed for hybrid and constrained MMG scheduling. The algorithm combines hybrid stochastic policy learning and twin-critic value estimation to handle mixed discrete-continuous dispatch decisions and improve cooperative learning stability, while EV-aware regularization and feasibility-guided action processing enhance mobility-reliability protection and physical consistency.
- (4) Comprehensive case studies on a real-world residential MMG system with mobile EV participation demonstrate that HEC-MADRL reduces the total objective value by 4.43% relative to the best-performing baseline, MAAC-PER, while improving low-carbon performance, operational feasibility, and EV mobility reliability with an online decision time of only 0.043 s.

1.4. Organization of this paper

The remainder of this paper is organized as follows. Section 2 presents the two-layer microgrid-EV mobility-energy coordination model. Section 3 reformulates the optimization problem as a cooperative multi-agent MDP. Section 4 introduces the proposed HEC-MADRL algorithm. Section 5 reports the case studies and performance analysis. Finally, Section 6 concludes this paper and discusses future research directions.

2. Problem formulation

The proposed framework is formulated as a two-layer decision-making architecture that integrates microgrid-side energy management with EV-side mobility-energy coordination, as illustrated in Fig. 1. The microgrid-side model characterizes the coordinated scheduling of dispatchable generation units, renewable energy resources, energy storage systems, demand response resources, grid power transactions, carbon emissions, and system-wide power balance. In parallel, the EV-side model describes mobility-dependent operational behaviors, including travel activities, station queuing, physical grid connection, charging and discharging processes, battery degradation, and range-anxiety effects. These two layers are interconnected through the aggregate EV charging and discharging power exchanged with each microgrid, thereby enabling coordinated optimization between microgrid operation and EV mobility-energy requirements. For ease of reference, the main indices, variables, and parameters used in the proposed formulation are summarized in Table 1.

The overall optimization objective is formulated as:

$$\min J = \omega_{MG} J_{MG} + \omega_{EV} J_{EV}, \quad (1)$$

where J_{MG} is the microgrid-side operating cost, J_{EV} is the EV-side user-related cost, and ω_{MG} and ω_{EV} are weighting coefficients that balance system-level operating performance and EV-user-related objectives.

2.1. Microgrid-side energy management model

Let $\mathcal{M}$ denote the set of microgrids, and let $\mathcal{T} = \{1, 2, \dots, T\}$ represent the set of scheduling intervals. For each microgrid $m \in \mathcal{M}$, $\mathcal{G}_m$ and $\mathcal{B}_m$ denote the set of dispatchable DGs and ESSs, respectively. The indices m , t , g , and b are used to indicate the microgrid, time interval, DG unit, and ESS unit, respectively. The duration of each scheduling interval is denoted by Δt .

The objective of the microgrid-side energy management model is to minimize the total operating cost over the scheduling horizon. This cost consists of the DG generation cost, grid transaction cost, ESS degradation cost, demand response compensation cost, and carbon emission cost, which can be expressed as:

$$J_{MG} = \sum_{t \in \mathcal{T}} \sum_{m \in \mathcal{M}} (C_{DG,m,t} + C_{ESS,m,t}^{deg} + C_{Grid,m,t} + C_{DR,m,t} + C_{CO_2,m,t}), \quad (2)$$

where $C_{DG,m,t}$ denotes the operating cost of dispatchable DGs, $C_{Grid,m,t}$ represents the grid transaction cost, $C_{ESS,m,t}^{deg}$ is the ESS degradation cost, $C_{DR,m,t}$ represents the compensation cost for demand response, and $C_{CO_2,m,t}$ denotes the carbon emission cost of microgrid m during scheduling interval t .

2.1.1. Dispatchable distributed generator model

Dispatchable DGs provide controllable generation resources to support reliable and economical microgrid operation. The operating cost of these units is represented by a quadratic cost function as follows:

$$C_{DG,m,t} = \sum_{g \in \mathcal{G}_m} [a_g (P_{DG,g,m,t})^2 + b_g P_{DG,g,m,t} + c_g] \Delta t, \quad (3)$$

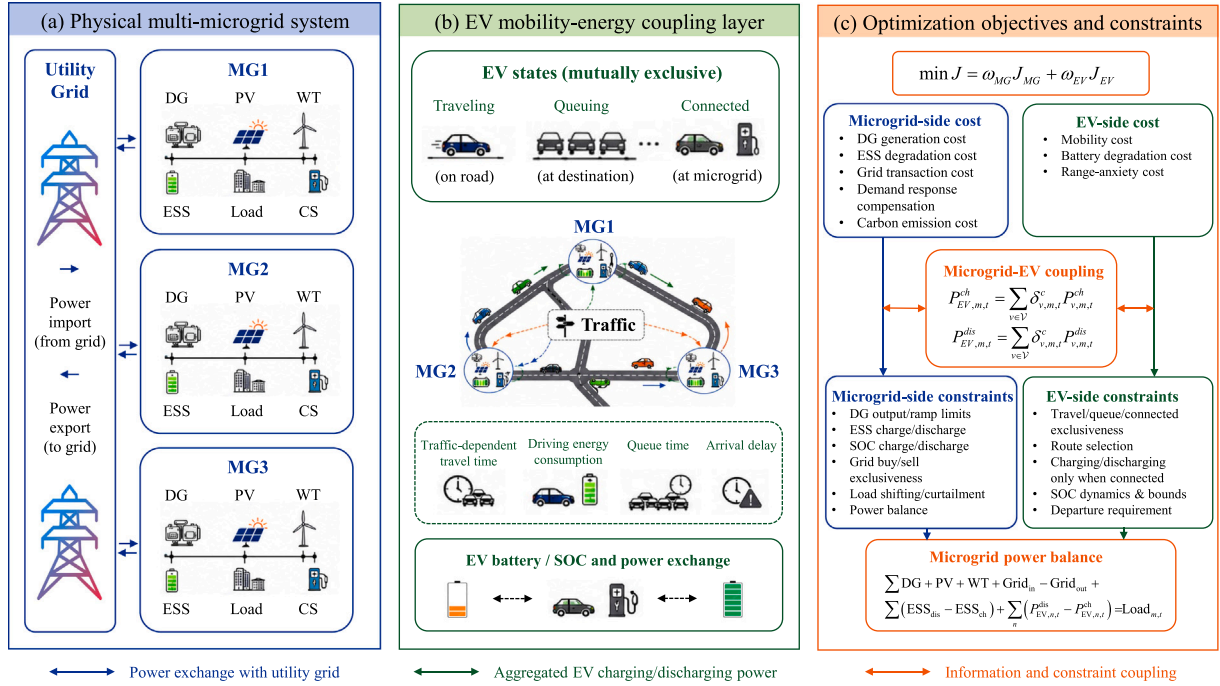


Fig. 1. Two-layer problem formulation for coordinated microgrid-EV mobility-energy management: (a) physical MMG system with DGs, PV units, WTs, ESSs, flexible loads, charging stations, and utility-grid interactions. (b) EV mobility-energy coupling layer, where EVs travel among microgrids and can exchange power only when physically connected to a charging station. (c) optimization objectives and coupling constraints, including microgrid-side operating costs, EV-side mobility-related costs, power-balance constraints, and aggregated EV charging/discharging power.

where $P_{DG,g,m,t}$ is the active power output of DG g in microgrid m at time t , and a_g , b_g , and c_g are the corresponding generation cost coefficients.

The active power output of each DG is constrained by its lower and upper generation limits,

$$P_{DG,g}^{\min} \leq P_{DG,g,m,t} \leq P_{DG,g}^{\max}, \quad \forall g \in \mathcal{G}_m, m \in \mathcal{M}, t \in \mathcal{T}, \quad (4)$$

where $P_{DG,g}^{\min}$ and $P_{DG,g}^{\max}$ are the minimum and maximum active power outputs of DG g , respectively.

To account for the temporal operating flexibility of dispatchable DGs, the inter-temporal ramping constraint is imposed as:

$$-R_{DG,g}^{\text{down}} \leq P_{DG,g,m,t} - P_{DG,g,m,t-1} \leq R_{DG,g}^{\text{up}}, \quad \forall g \in \mathcal{G}_m, m \in \mathcal{M}, t \in \mathcal{T}, \quad (5)$$

where $R_{DG,g}^{\text{up}}$ and $R_{DG,g}^{\text{down}}$ represent the upward and downward ramping limits of DG g , respectively.

2.1.2. Photovoltaic generation model

PV generation is regarded as a non-dispatchable renewable source, whose output is primarily governed by solar irradiance and temperature-dependent conversion characteristics. For microgrid m during scheduling interval t , the PV power output is expressed as

$$P_{PV,m,t} = P_{PV,m}^r \frac{G_{m,t}}{1000} [1 + \theta_{PV} (T_{PV,m,t} - T_c)], \quad (6)$$

where $P_{PV,m}^r$ denotes the rated PV capacity installed in microgrid m , $G_{m,t}$ is the solar irradiance, θ_{PV} is the temperature coefficient of the PV module, $T_{PV,m,t}$ is the actual PV module temperature, and T_c is the rated reference temperature, which is typically set to 25 °C.

Table 1

Nomenclature.

Symbol	Description
Indices and Sets	
m, t, g, b, v, k, a	Indices of microgrid, scheduling interval, DG unit, ESS unit, EV, route, and road link, respectively.
$\mathcal{M}, \mathcal{T}$	Sets of microgrids and scheduling intervals, respectively.
$\mathcal{G}_m, \mathcal{B}_m$	Sets of dispatchable DGs and ESSs in microgrid m , respectively.
$\mathcal{V}, \mathcal{K}_v, \mathcal{A}'$	Sets of EVs, feasible routes of EV v , and road links, respectively.
Δt	Duration of each scheduling interval.
Objective and Cost Terms	
J, J_{MG}, J_{EV}	Overall objective, microgrid-side operating cost, and EV-side user-related cost, respectively.
$C_{DG,m,t}, C_{Grid,m,t}$	DG operating cost and grid transaction cost of microgrid m during interval t , respectively.
$C_{ESS,m,t}^{\text{deg}}, C_{DR,m,t}, C_{CO_2,m,t}$	ESS degradation cost, demand response compensation cost, and carbon emission cost, respectively.
$C_{EV,t}^{\text{mob}}, C_{EV,t}^{\text{deg}}, C_{EV,t}^{\text{risk}}$	EV mobility-related cost, EV battery degradation cost, and range-anxiety cost during interval t , respectively.
$C_{route,v,t}, C_{detour,v,t}, C_{queue,v,t}, C_{delay,v,t}$	Route-related, detour, queuing, and terminal arrival-delay costs of EV v , respectively.
Microgrid-Side Variables	
$P_{DG,g,m,t}$	Active power output of DG g in microgrid m during interval t .
$P_{DG,g}^{\min}, P_{DG,g}^{\max}$	Minimum and maximum active power outputs of DG g , respectively.
$P_{PV,m,t}, P_{PV,m}^r$	PV power output and rated PV capacity of microgrid m , respectively.
$G_{m,t}, T_{PV,m,t}$	Solar irradiance and PV module temperature of microgrid m during interval t , respectively.

(continued on next page)

Table 1 (continued)

Symbol	Description
$P_{WT,m,t}, P_{WT,m}^r$	WT power output and rated WT capacity of microgrid m , respectively.
$v_{m,t}$	Wind speed of microgrid m during interval t .
$P_{b,m,t}^{ch}, P_{b,m,t}^{dis}$	Charging and discharging powers of ESS b in microgrid m , respectively.
$u_{b,m,t}^{ch}, u_{b,m,t}^{dis}$	Binary variables indicating the charging and discharging states of ESS b , respectively.
$SOC_{b,m,t}$	State of charge of ESS b in microgrid m during interval t .
E_b	Rated energy capacity of ESS b .
$P_{Grid,m,t}^{buy}, P_{Grid,m,t}^{sell}$	Power purchased from and sold to the utility grid by microgrid m , respectively.
$u_{Grid,m,t}^{buy}, u_{Grid,m,t}^{sell}$	Binary variables indicating grid-buying and grid-selling states, respectively.
$P_{L,m,t}$	Actual load of microgrid m after demand response.
$P_{fix,m,t}, P_{shift,m,t}, P_{cur,m,t}$	Fixed load, shifted load, and curtailed load of microgrid m , respectively.
$P_{base,m,t}$	Baseline load of microgrid m during interval t .
$E_{CO_2,m,t}$	Carbon emissions of microgrid m during interval t .
EV-Side Mobility Variables	
$\delta_{v,t}^{it}$	Binary variable indicating whether EV v is traveling during interval t .
$\delta_{v,m,t}^q, \delta_{v,m,t}^c$	Binary variables indicating whether EV v is queuing at or physically connected to microgrid m , respectively.
$y_{v,m,t}$	Access status of EV v at microgrid m , indicating whether it is queuing or physically connected.
$x_{v,k,t}$	Binary variable indicating whether EV v selects route k during interval t .
$T_{v,k,t}^r, E_{v,k,t}^r$	Travel time and driving energy consumption of EV v on route k , respectively.
$\tau_{a,t}, q_{a,t}, Q_a^{max}$	Travel time, traffic flow, and capacity of road link a , respectively.
$D_{v,m,t}^{act}, D_{v,t}^{ori}$	Actual travel distance after accessing microgrid m and original shortest travel distance of EV v , respectively.
$Q_{m,t}$	Expected queuing time at microgrid m during interval t .
$t_v^{arr}, t_v^{arr,des}$	Actual and desired arrival times of EV v , respectively.
EV-Side Energy Variables	
$P_{v,m,t}^{ch}, P_{v,m,t}^{dis}$	Charging and discharging powers of EV v connected to microgrid m , respectively.
$P_{EV,m,t}^{ch}, P_{EV,m,t}^{dis}$	Aggregated EV charging and discharging powers exchanged with microgrid m , respectively.
$SOC_{v,t}$	State of charge of EV v during interval t .
SOC_v^{req}, SOC_v^{res}	Required SOC and reserved SOC margin of EV v , respectively.
E_v	Rated battery capacity of EV v .
C_v^{rep}, N_v^{cyc}	Battery replacement cost and expected cycle life of EV v , respectively.
t_v^{dep}	Departure time of EV v .
$\bar{E}_v^{trip}, D_v^{trip}$	Predicted energy demand and expected distance of the upcoming trip of EV v , respectively.
$R_{v,t}^{risk}$	Range-anxiety risk of EV v during interval t .

2.1.3. Wind turbine generation model

WT generation is modeled using a standard wind-speed-dependent power curve:

$$P_{WT,m,t} = \begin{cases} 0, & v_{m,t} < v_{ci} \text{ or } v_{m,t} > v_{co}, \\ P_{WT,m}^r \frac{v_{m,t} - v_{ci}}{v_r - v_{ci}}, & v_{ci} \leq v_{m,t} < v_r, \\ P_{WT,m}^r, & v_r \leq v_{m,t} \leq v_{co}. \end{cases} \quad (7)$$

Accordingly, the WT output is bounded by its rated capacity as:

$$0 \leq P_{WT,m,t} \leq P_{WT,m}^r, \quad \forall m \in \mathcal{M}, t \in \mathcal{T}, \quad (8)$$

where $P_{WT,m,t}$ denotes the WT power output of microgrid m at time t , $v_{m,t}$ is the corresponding wind speed, and v_{ci} , v_r , and v_{co} represent the

cut-in, rated, and cut-out wind speeds, respectively. $P_{WT,m}^r$ denotes the rated WT capacity installed in microgrid m .

2.1.4. Energy storage system model

ESSs are incorporated to mitigate renewable generation variability and improve the operational flexibility of each microgrid. For ESS b in microgrid m , the charging and discharging powers are constrained by its rated power limits as follows:

$$0 \leq P_{b,m,t}^{ch} \leq u_{b,m,t}^{ch} P_b^{ch,max}, \quad \forall b \in \mathcal{B}_m, m \in \mathcal{M}, t \in \mathcal{T}, \quad (9)$$

$$0 \leq P_{b,m,t}^{dis} \leq u_{b,m,t}^{dis} P_b^{dis,max}, \quad \forall b \in \mathcal{B}_m, m \in \mathcal{M}, t \in \mathcal{T}, \quad (10)$$

$$u_{b,m,t}^{ch} + u_{b,m,t}^{dis} \leq 1, u_{b,m,t}^{ch}, u_{b,m,t}^{dis} \in \{0, 1\}, \quad (11)$$

$$\forall b \in \mathcal{B}_m, m \in \mathcal{M}, t \in \mathcal{T},$$

where P_b^{ch} and P_b^{dis} denote the charging and discharging powers of ESS b , respectively. $P_b^{ch,max}$ and $P_b^{dis,max}$ are the corresponding maximum charging and discharging power limits. The binary variables $u_{b,m,t}^{ch}$ and $u_{b,m,t}^{dis}$ indicate the charging and discharging states, respectively. Constraint (11) ensures that an ESS cannot be charged and discharged simultaneously.

The SOC evolution of ESS b is described by

$$SOC_{b,m,t+1} = SOC_{b,m,t} + \frac{\eta_b^{ch} P_{b,m,t}^{ch} \Delta t}{E_b} - \frac{P_{b,m,t}^{dis} \Delta t}{\eta_b^{dis} E_b}, \quad (12)$$

$$\forall b \in \mathcal{B}_m, m \in \mathcal{M}, t \in \mathcal{T},$$

where $SOC_{b,m,t}$ is the state of charge of ESS b , E_b denotes its rated energy capacity, and η_b^{ch} and η_b^{dis} are the charging and discharging efficiencies, respectively. The SOC is further bounded within its admissible operating range:

$$SOC_b^{min} \leq SOC_{b,m,t} \leq SOC_b^{max}, \quad \forall b \in \mathcal{B}_m, m \in \mathcal{M}, t \in \mathcal{T}, \quad (13)$$

where SOC_b^{min} and SOC_b^{max} denote the minimum and maximum allowable SOC levels.

The degradation cost of ESSs is modeled as a linear function of energy throughput, i.e.,

$$C_{ESS,m,t}^{deg} = \sum_{b \in \mathcal{B}_m} c_b^{deg} (P_{b,m,t}^{ch} + P_{b,m,t}^{dis}) \Delta t, \quad (14)$$

where c_b^{deg} represents the unit degradation cost associated with the charging and discharging energy throughput of ESS b .

2.1.5. Utility grid interaction model

Each microgrid is allowed to exchange electricity with the upstream utility grid through power import and export transactions. The corresponding grid transaction cost is formulated as:

$$C_{Grid,m,t} = \rho_t^{buy} P_{Grid,m,t}^{buy} \Delta t - \rho_t^{sell} P_{Grid,m,t}^{sell} \Delta t, \quad (15)$$

where $P_{Grid,m,t}^{buy}$ and $P_{Grid,m,t}^{sell}$ denote the power purchased from and sold to the utility grid by microgrid m during interval t , respectively. In addition, ρ_t^{buy} and ρ_t^{sell} represent the time-varying electricity buying and selling prices, respectively.

The grid import and export power are limited by the interconnection capacity as follows:

$$0 \leq P_{Grid,m,t}^{buy} \leq u_{Grid,m,t}^{buy} P_{Grid,m}^{buy,max}, \quad \forall m \in \mathcal{M}, t \in \mathcal{T}, \quad (16)$$

$$0 \leq P_{Grid,m,t}^{sell} \leq u_{Grid,m,t}^{sell} P_{Grid,m}^{sell,max}, \quad \forall m \in \mathcal{M}, t \in \mathcal{T}, \quad (17)$$

$$u_{Grid,m,t}^{buy} + u_{Grid,m,t}^{sell} \leq 1, u_{Grid,m,t}^{buy}, u_{Grid,m,t}^{sell} \in \{0, 1\}, \quad (18)$$

$$\forall m \in \mathcal{M}, t \in \mathcal{T},$$

where $P_{Grid,m}^{buy,max}$ and $P_{Grid,m}^{sell,max}$ denote the maximum import and export capacities of microgrid m , respectively. The binary variables $u_{Grid,m,t}^{buy}$

and $u_{Grid,m,t}^{sell}$ indicate the grid-buying and grid-selling states. Constraint (18) enforces the mutually exclusive operation of grid import and export, thereby preventing simultaneous electricity purchasing and selling within the same scheduling interval.

2.1.6. Demand response model

DR is incorporated to exploit load-side flexibility and enhance the economic operation of microgrids. After DR implementation, the actual load of microgrid m during scheduling interval t is given by

$$P_{L,m,t} = P_{fix,m,t} + P_{shift,m,t} - P_{cur,m,t}, \quad (19)$$

where $P_{L,m,t}$ denotes the actual load after DR, $P_{fix,m,t}$ is the fixed load, $P_{shift,m,t}$ represents the shifted load, and $P_{cur,m,t}$ is the curtailed load. A positive value of $P_{shift,m,t}$ indicates load shifted into interval t , whereas a negative value indicates load shifted out of interval t .

The curtailed load is limited by the allowable curtailment ratio:

$$0 \leq P_{cur,m,t} \leq \mu_{cur} P_{base,m,t}, \quad \forall m \in \mathcal{M}, t \in \mathcal{T}, \quad (20)$$

where $P_{base,m,t}$ denotes the baseline load, and μ_{cur} is the maximum allowable load curtailment ratio.

Similarly, the shifted load is constrained within the prescribed flexibility range:

$$-\mu_{shift} P_{base,m,t} \leq P_{shift,m,t} \leq \mu_{shift} P_{base,m,t}, \quad \forall m \in \mathcal{M}, t \in \mathcal{T}, \quad (21)$$

where μ_{shift} denotes the maximum load shifting ratio. To ensure that load shifting only redistributes demand over time rather than changing the total energy consumption, the following energy conservation constraint is imposed:

$$\sum_{t \in \mathcal{T}} P_{shift,m,t} \Delta t = 0, \quad \forall m \in \mathcal{M}. \quad (22)$$

The DR compensation cost is modeled as

$$C_{DR,m,t} = c_{cur} P_{cur,m,t} \Delta t + c_{shift} |P_{shift,m,t}| \Delta t, \quad (23)$$

where c_{cur} and c_{shift} are the compensation coefficients associated with load curtailment and load shifting, respectively. This formulation enables the microgrid to utilize demand-side flexibility while explicitly accounting for the economic compensation required for load adjustment.

2.1.7. Carbon emission model

To encourage low-carbon microgrid operation, carbon emissions associated with dispatchable DG generation and electricity purchased from the utility grid are incorporated into the objective function. The carbon emissions of microgrid m during scheduling interval t are calculated as

$$E_{CO_2,m,t} = \sum_{g \in \mathcal{G}_m} \gamma_{DG,g} P_{DG,g,m,t} \Delta t + \gamma_{Grid,t} P_{Grid,m,t}^{buy} \Delta t, \quad (24)$$

where $\gamma_{DG,g}$ denotes the carbon emission factor of DG g , and $\gamma_{Grid,t}$ represents the time-dependent carbon emission factor associated with electricity purchased from the utility grid. The corresponding carbon emission cost is then expressed as:

$$C_{CO_2,m,t} = \rho_{CO_2} E_{CO_2,m,t}, \quad (25)$$

where ρ_{CO_2} denotes the carbon price.

2.1.8. Microgrid power balance

For each microgrid and scheduling interval, the power balance constraint ensures that the total available supply equals the total demand.

Specifically, the balance for microgrid m during interval t is formulated as

$$\begin{aligned} & \sum_{g \in \mathcal{G}_m} P_{DG,g,m,t} + P_{PV,m,t} + P_{WT,m,t} + P_{Grid,m,t}^{buy} - P_{Grid,m,t}^{sell} \\ & + \sum_{b \in \mathcal{B}_m} (P_{b,m,t}^{dis} - P_{b,m,t}^{ch}) + P_{EV,m,t}^{dis} - P_{EV,m,t}^{ch} = P_{L,m,t}, \end{aligned} \quad (26)$$

where $P_{EV,m,t}^{ch}$ and $P_{EV,m,t}^{dis}$ denote the aggregated charging and discharging powers of EVs connected to microgrid m during interval t , respectively. This constraint couples the microgrid-side energy management model with the EV-side mobility-energy model by explicitly incorporating the net EV power exchange into the microgrid power balance.

2.2. EV-side mobility-energy coupling model

The EV-side model characterizes the spatio-temporal mobility of EVs across multiple microgrids and links individual EV charging/discharging decisions to the microgrid-level power balance through aggregate interface variables. The EV-side objective is formulated as:

$$J_{EV} = \sum_{t \in \mathcal{T}} (C_{EV,t}^{mob} + C_{EV,t}^{deg} + C_{EV,t}^{risk}), \quad (27)$$

where $C_{EV,t}^{mob}$, $C_{EV,t}^{deg}$, and $C_{EV,t}^{risk}$ denote the EV mobility-related cost, battery degradation cost, and range-anxiety cost during interval t , respectively.

In the considered MMG system, EVs may travel among different microgrids or charging stations. Energy exchange between an EV and a microgrid is permitted only when the EV is physically connected to a charger. Accordingly, each EV is assigned one of three mutually exclusive operating states, namely traveling, queuing, or connected. The operating status of each EV is characterized by three binary variables, namely $\delta_{v,t}^{tr}$, $\delta_{v,m,t}^q$, and $\delta_{v,m,t}^c$. Specifically, $\delta_{v,t}^{tr} = 1$ indicates that EV v is traveling during scheduling interval t , and $\delta_{v,t}^{tr} = 0$ otherwise. Similarly, $\delta_{v,m,t}^q = 1$ indicates that EV v is waiting in the queue at microgrid m during interval t , and $\delta_{v,m,t}^q = 0$ otherwise. The physical connection status is represented by $\delta_{v,m,t}^c$, where $\delta_{v,m,t}^c = 1$ means that EV v is physically connected to a charger in microgrid m during interval t , and $\delta_{v,m,t}^c = 0$ otherwise. These binary variables jointly describe the mobility and charging-access status of each EV across the scheduling horizon.

The mutually exclusive state condition is imposed as

$$\delta_{v,t}^{tr} + \sum_{m \in \mathcal{M}} \delta_{v,m,t}^q + \sum_{m \in \mathcal{M}} \delta_{v,m,t}^c = 1, \quad \forall v \in \mathcal{V}, t \in \mathcal{T}. \quad (28)$$

This constraint ensures that, in each scheduling interval, every EV can be in only one of the following states: traveling, queuing at one microgrid, or physically connected to one microgrid.

For notational compactness, the access status of EV v at microgrid m is further defined as

$$y_{v,m,t} = \delta_{v,m,t}^q + \delta_{v,m,t}^c. \quad (29)$$

Thus, $y_{v,m,t} = 1$ indicates that EV v has arrived at microgrid m , either waiting in the queue or physically connected to a charger.

2.2.1. EV mobility cost model

The EV mobility cost is introduced to quantify the user-side inconvenience caused by energy-related routing and charging decisions. It consists of route travel time, driving energy consumption, detour cost, queuing cost, and arrival delay cost. Although this cost term does not directly enter the microgrid power balance, it determines the spatio-temporal availability of EVs at different microgrids through the connection status $\delta_{v,m,t}^c$ and the access status $y_{v,m,t}$. Consequently, it affects the aggregated EV charging and discharging powers, i.e., $P_{EV,m,t}^{ch}$

and $P_{EV,m,t}^{dis}$, thereby coupling EV mobility behavior with microgrid-side energy management.

Let $x_{v,k,t}$ be a binary route-selection variable, where $x_{v,k,t} = 1$ indicates that EV v selects route k during scheduling interval t , and $x_{v,k,t} = 0$ otherwise. The route-selection constraint is given by

$$\sum_{k \in \mathcal{K}_v} x_{v,k,t} = \delta_{v,t}^{tr}, \quad \forall v \in \mathcal{V}, t \in \mathcal{T}, \quad (30)$$

which ensures that each EV selects exactly one feasible route only when it is in the traveling state.

The travel time of EV v on route k is expressed as

$$T_{v,k,t}^r = \sum_{a \in \mathcal{A}^r} \chi_{a,k} \tau_{a,t}, \quad (31)$$

where $\chi_{a,k}$ is a binary parameter indicating whether road link a belongs to route k , and $\tau_{a,t}$ denotes the travel time of road link a during interval t . The link travel time is described by the Bureau of Public Roads (BPR) [32] function:

$$\tau_{a,t} = \tau_a^0 \left[1 + \alpha_a \left(\frac{q_{a,t}}{Q_a^{\max}} \right)^{\beta_a} \right], \quad (32)$$

where τ_a^0 is the free-flow travel time, $q_{a,t}$ is the traffic flow, $Q_a^{\max}$ is the road capacity, and α_a and β_a are congestion sensitivity coefficients.

The driving energy consumption of EV v on route k is calculated as

$$E_{v,k,t}^r = \sum_{a \in \mathcal{A}^r} \chi_{a,k} e_{v,a,t}, \quad (33)$$

where $e_{v,a,t}$ denotes the driving energy consumption of EV v on road link a . Specifically, it is modeled as

$$e_{v,a,t} = \left(\varphi_{v,0} + \varphi_{v,1} u_{a,t} + \varphi_{v,2} u_{a,t}^2 + \varphi_{v,3} \zeta_a \right) l_a, \quad (34)$$

where $u_{a,t}$ is the average speed on link a , ζ_a is the road gradient, l_a is the link length, and $\varphi_{v,0}$, $\varphi_{v,1}$, $\varphi_{v,2}$, and $\varphi_{v,3}$ are EV-specific energy-consumption coefficients.

Accordingly, the route-related mobility cost of EV v is formulated as

$$C_{v,t}^{route} = \lambda_T \sum_{k \in \mathcal{K}_v} x_{v,k,t} T_{v,k,t}^r + \lambda_E \sum_{k \in \mathcal{K}_v} x_{v,k,t} E_{v,k,t}^r, \quad (35)$$

where λ_T denotes the value-of-time coefficient, and λ_E is the unit driving energy cost coefficient.

When EV v deviates from its original route to access a microgrid for energy interaction, the corresponding detour cost is defined as

$$C_{v,t}^{detour} = \lambda_D \sum_{m \in \mathcal{M}} y_{v,m,t} \max \left(0, D_{v,m,t}^{act} - D_{v,t}^{ori} \right), \quad (36)$$

where $D_{v,m,t}^{act}$ is the actual travel distance after EV v accesses microgrid m , $D_{v,t}^{ori}$ is the original shortest travel distance without energy-related detour, and λ_D is the unit detour cost coefficient.

The queuing cost is determined by the queuing status and the expected waiting time at each microgrid:

$$C_{v,t}^{queue} = \lambda_Q \sum_{m \in \mathcal{M}} \delta_{v,m,t}^q Q_{m,t}, \quad (37)$$

where $Q_{m,t}$ is the expected queuing time at microgrid m , and λ_Q denotes the waiting-time cost coefficient.

The arrival delay cost is expressed as

$$C_{v,t}^{delay} = \lambda_{delay} \max \left(0, t_v^{arr} - t_v^{arr,des} \right), \quad (38)$$

where t_v^{arr} and $t_v^{arr,des}$ denote the actual and desired arrival times of EV v , respectively.

Therefore, the total EV mobility cost during interval t is given by

$$C_{EV,t}^{mob} = \sum_{v \in \mathcal{V}} \left(C_{v,t}^{route} + C_{v,t}^{detour} + C_{v,t}^{queue} + C_{v,t}^{delay} \right). \quad (39)$$

The aggregated EV charging and discharging power exchanged with microgrid m are further defined as

$$P_{EV,m,t}^{ch} = \sum_{v \in \mathcal{V}} \delta_{v,m,t}^c P_{v,m,t}^{ch}, \quad \forall m \in \mathcal{M}, t \in \mathcal{T}, \quad (40)$$

$$P_{EV,m,t}^{dis} = \sum_{v \in \mathcal{V}} \delta_{v,m,t}^c P_{v,m,t}^{dis}, \quad \forall m \in \mathcal{M}, t \in \mathcal{T}. \quad (41)$$

Eqs. (40) and (41) establish the interface between the EV-side mobility model and the microgrid-side power balance. Specifically, only EVs physically connected to microgrid m can participate in charging or discharging, and their aggregated power exchange directly affects the operation of the corresponding microgrid.

2.2.2. EV battery degradation model

EV battery degradation is explicitly considered to account for the aging effect induced by frequent charging and discharging operations, particularly under V2G participation. Since an EV can exchange power with a microgrid only when it is physically connected to a charger, the charging and discharging powers are constrained by the connection status as follows:

$$0 \leq P_{v,m,t}^{ch} \leq \delta_{v,m,t}^c P_v^{ch,max}, \quad \forall v \in \mathcal{V}, m \in \mathcal{M}, t \in \mathcal{T}, \quad (42)$$

$$0 \leq P_{v,m,t}^{dis} \leq \delta_{v,m,t}^c P_v^{dis,max}, \quad \forall v \in \mathcal{V}, m \in \mathcal{M}, t \in \mathcal{T}. \quad (43)$$

These constraints ensure that EV v can participate in charging or discharging only when it is connected to microgrid m ; otherwise, its charging and discharging powers are forced to zero during traveling or queuing states.

Based on the energy-throughput approximation, the degradation cost incurred by EV v when connected to microgrid m during interval t is modeled as

$$C_{v,m,t}^{deg} = c_v^{deg} \left(P_{v,m,t}^{ch} + P_{v,m,t}^{dis} \right) \Delta t. \quad (44)$$

The unit degradation cost coefficient is calculated as

$$c_v^{deg} = \frac{C_v^{rep}}{2N_v^{cyc} E_v}, \quad (45)$$

where C_v^{rep} denotes the battery replacement cost of EV v , N_v^{cyc} is the expected cycle life, and E_v is the rated battery capacity. The denominator $2N_v^{cyc} E_v$ represents the total lifetime energy throughput under full charge-discharge cycling.

Accordingly, the total EV battery degradation cost during interval t is given by

$$C_{EV,t}^{deg} = \sum_{v \in \mathcal{V}} \sum_{m \in \mathcal{M}} c_v^{deg} \left(P_{v,m,t}^{ch} + P_{v,m,t}^{dis} \right) \Delta t. \quad (46)$$

This formulation provides a tractable representation of EV battery aging and discourages excessive charging/discharging actions in the coordinated mobility-energy management process.

2.2.3. EV range-anxiety model

Range anxiety reflects the concerns of EV users regarding whether the remaining battery energy is sufficient to complete subsequent trips. In an MMG environment, excessive V2G discharging may improve system-level operational performance but may compromise the mobility reliability of EV users. Therefore, the EV SOC dynamics and departure energy requirement are explicitly incorporated into the model.

The SOC evolution of EV v is formulated as

$$SOC_{v,t+1} = SOC_{v,t} + \sum_{m \in \mathcal{M}} \frac{\eta_v^{ch} P_{v,m,t}^{ch} \Delta t}{E_v} - \sum_{m \in \mathcal{M}} \frac{P_{v,m,t}^{dis} \Delta t}{\eta_v^{dis} E_v} - \frac{E_{v,t}^r}{E_v}, \quad \forall v \in \mathcal{V}, t \in \mathcal{T}, \quad (47)$$

where $SOC_{v,t}$ denotes the SOC of EV v at interval t , E_v is its rated battery capacity, and η_v^{ch} and η_v^{dis} are the charging and discharging efficiencies, respectively. The term $E_{v,t}^r$ represents the driving energy consumed during interval t . Since $P_{v,m,t}^{ch}$ and $P_{v,m,t}^{dis}$ are constrained by the connection status $\delta_{v,m,t}^c$, only the microgrid to which EV v is physically connected can contribute to its charging or discharging process.

The SOC of each EV is restricted to its allowable operating range:

$$SOC_v^{\min} \leq SOC_{v,t} \leq SOC_v^{\max}, \quad \forall v \in \mathcal{V}, t \in \mathcal{T}. \quad (48)$$

To guarantee mobility reliability, each EV is required to maintain a sufficient SOC before departure:

$$SOC_{v,t}^{dep} \geq SOC_v^{req}, \quad \forall v \in \mathcal{V}. \quad (49)$$

The required SOC is determined by a reserved SOC margin and the predicted energy demand for the upcoming trip:

$$SOC_v^{req} = SOC_v^{res} + \frac{\hat{E}_v^{trip}}{E_v}, \quad (50)$$

where SOC_v^{res} denotes the reserve SOC margin, and $\hat{E}_v^{trip}$ is the estimated energy consumption of the subsequent trip. The latter is calculated as

$$\hat{E}_v^{trip} = e_v^{mile} D_v^{trip}, \quad (51)$$

where e_v^{mile} is the unit-distance energy consumption, and D_v^{trip} denotes the expected travel distance.

The range-anxiety risk is defined as the SOC deficiency relative to the required SOC:

$$R_{v,t}^{risk} = \max(0, SOC_v^{req} - SOC_{v,t}). \quad (52)$$

Accordingly, the range-anxiety cost of EV v is modeled as a quadratic penalty:

$$C_{v,t}^{risk} = \lambda_{risk} (R_{v,t}^{risk})^2, \quad (53)$$

where λ_{risk} is the penalty coefficient for range-anxiety risk.

The total range-anxiety cost of all EVs during interval t is then given by

$$C_{EV,t}^{risk} = \sum_{v \in \mathcal{V}} \lambda_{risk} [\max(0, SOC_v^{req} - SOC_{v,t})]^2. \quad (54)$$

This formulation penalizes insufficient SOC levels and prevents the coordinated energy management strategy from overusing EV batteries for V2G services at the expense of users' mobility requirements.

3. Multi-agent MDP formulation

The optimization problem in Section 2 is inherently sequential, as current scheduling decisions affect subsequent DG outputs, ESS and EV SOC trajectories, operating costs, carbon emissions, and mobility reliability. Given the geographically distributed nature of microgrids and the mobility of EVs, a centralized single-agent formulation may suffer from poor scalability. Therefore, the coupled microgrid-EV energy management problem is reformulated as a cooperative multi-agent MDP, where each microgrid acts as an autonomous learning agent.

In this formulation, EVs are modeled as mobile flexible resources rather than independent agents. Their traveling, queuing, and physical connection states are incorporated into the environment, while the charging and discharging powers of physically connected EVs are determined by the corresponding microgrid agent. This design is aligned with the energy-management-oriented objective of coordinating microgrid operation and EV mobility-energy behavior.

Fig. 2 illustrates the proposed multi-agent MDP reformulation of the coupled microgrid-EV scheduling problem. The framework converts the original deterministic optimization model into a closed-loop sequential decision-making process, where global system information is transformed into local observations, decentralized microgrid agents generate local scheduling actions, and the coupled environment updates the system state and cooperative reward. The cooperative multi-agent MDP is defined as

$$\mathcal{G}_{MDP} = (\mathcal{M}, S, \{\mathcal{O}_m\}_{m \in \mathcal{M}}, \{\mathcal{A}_m\}_{m \in \mathcal{M}}, \mathcal{P}, \mathcal{R}, \gamma), \quad (55)$$

where S denotes the global state space, $\mathcal{O}_m$ and $\mathcal{A}_m$ are the local observation and action spaces of agent m , respectively, $\mathcal{P}$ is the state transition function, $\mathcal{R}$ is the cooperative reward function, and $\gamma \in [0, 1]$ is the discount factor. Each agent learns a decentralized policy as $a_{m,t} \sim \pi_{\theta_m}(a_{m,t} | o_{m,t})$, where θ_m denotes the parameters of the policy network of agent m .

3.1. Global state

At each scheduling interval t , the global state aggregates the operating information of all microgrids and the mobility-energy status of EVs, and is defined as

$$s_t = \left\{ \left\{ s_{m,t}^{MG} \right\}_{m \in \mathcal{M}}, s_t^{EV} \right\}. \quad (56)$$

The microgrid-side state component of agent m is given by

$$s_{m,t}^{MG} = [P_{PV,m,t}, P_{WT,m,t}, P_{L,m,t}, \{SOC_{b,m,t}\}_{b \in \mathcal{B}_m}, \{P_{DG,g,m,t-1}\}_{g \in \mathcal{G}_m}, \rho_t^{buy}, \rho_t^{sell}, \Xi_{m,t}^{shift}]. \quad (57)$$

Most variables in Eq. (57) have been defined in Section 2. The previous DG output $P_{DG,g,m,t-1}$ is included to preserve the Markov property under the ramping constraint in Eq. (5). In addition, $\Xi_{m,t}^{shift}$ is introduced as an auxiliary state variable to record the cumulative shifted-load energy prior to interval t , thereby capturing the intertemporal demand-response constraint in Eq. (22):

$$\Xi_{m,t}^{shift} = \sum_{\tau=1}^{t-1} P_{shift,m,\tau} \Delta t. \quad (58)$$

The EV-side state component is formulated as

$$s_t^{EV} = [SOC_{v,t}, \delta_{v,t}^{tr}, \delta_{v,m,t}^q, \delta_{v,m,t}^c, y_{v,m,t}, \{x_{v,k,t}\}_{k \in \mathcal{K}_v}, Q_{m,t}, t_v^{arr}, t_v^{arr,des}, \tau_{v,t}^{dep}, D_v^{trip}, SOC_v^{req}, R_{v,t}^{risk}, E_{v,t}^r]_{v \in \mathcal{V}, m \in \mathcal{M}}. \quad (59)$$

The state variables in Eq. (59) are inherited from the EV-side mobility-energy model in Section 2, except for the remaining time to departure, which is defined as

$$\tau_{v,t}^{dep} = t_v^{dep} - t. \quad (60)$$

This temporal feature provides the agent with explicit departure-time information and facilitates policy learning under time-coupled EV mobility and charging requirements.

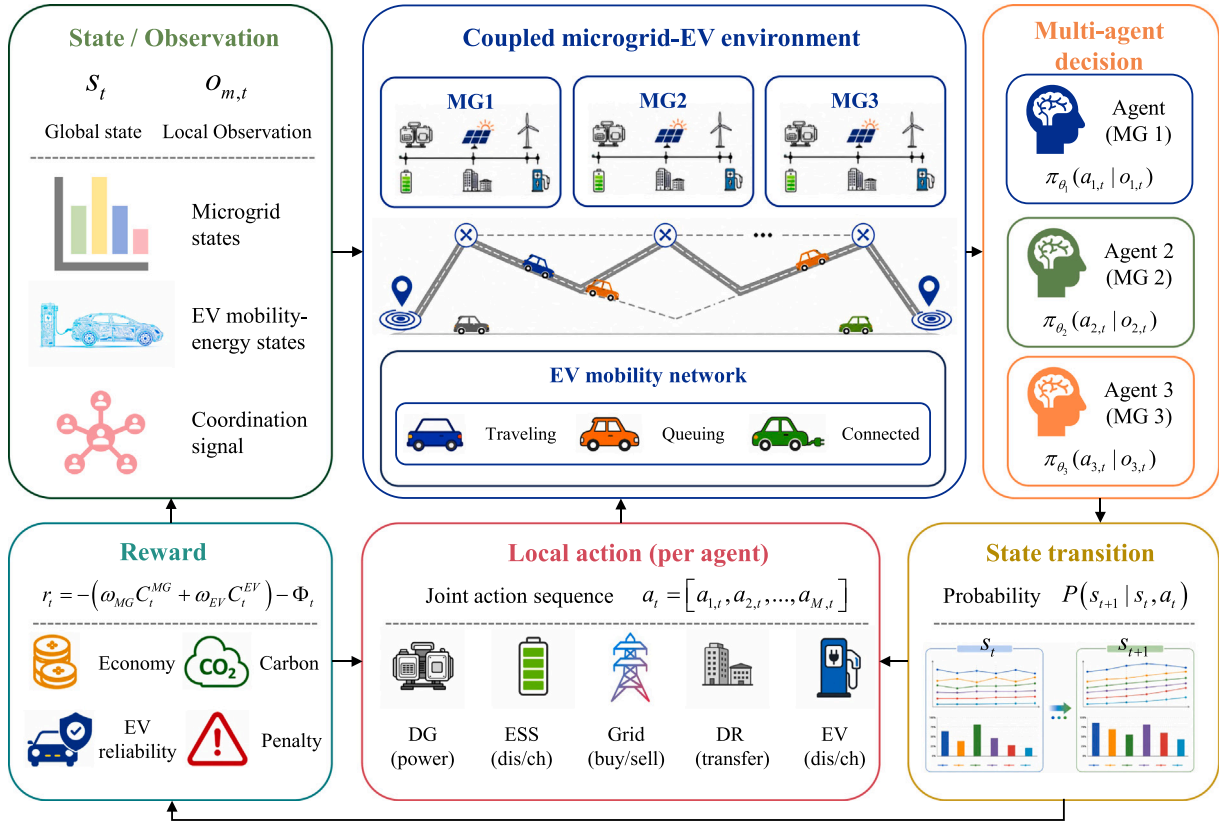


Fig. 2. Multi-agent MDP framework for coordinated microgrid-EV mobility-energy management. Each microgrid agent observes local microgrid information, EV mobility-energy states, and compact coordination signals, and then generates local scheduling actions for DGs, ESSs, grid transactions, demand response, and connected EVs. The aggregated joint action is executed in the coupled microgrid-EV environment, which updates the next state through power-balance constraints, SOC dynamics, EV mobility transitions, renewable generation, load variation, and electricity-price changes. The cooperative reward feeds back economic, carbon-emission, EV-reliability, and constraint-violation information for sequential policy learning.

3.2. Local observation of each microgrid agent

During decentralized execution, each microgrid agent makes decisions based on its local operating information and limited coordination signals from other microgrids. The local observation of agent m at interval t is defined as

$$o_{m,t} = [s_{m,t}^{MG}, \psi_{m,t}^{EV}, s_{m,t}^{sys}], \quad (61)$$

where $s_{m,t}^{MG}$ denotes the local microgrid state, $\psi_{m,t}^{EV}$ is an attention-based representation of the EVs associated with microgrid m , and $s_{m,t}^{sys}$ is a compact inter-microgrid coordination signal.

Let

$$\mathcal{V}_m^a(t) = \{v \in \mathcal{V} \mid y_{v,m,t} = 1\} \quad (62)$$

denote the set of EVs that have arrived at microgrid m , including both queuing and physically connected EVs. For each $v \in \mathcal{V}_m^a(t)$, the EV feature vector is constructed as

$$e_{v,m,t} = [SOC_{v,t}, \delta_{v,m,t}^a, \delta_{v,m,t}^c, Q_{m,t}, SOC_{v,t}^{req}, R_{v,t}^{risk}, D_v^{trip}, t_v^{arr}, t_v^{arr,des}, \tau_{v,t}^{dep}]. \quad (63)$$

The feature vector is then mapped into a latent representation through a learnable nonlinear embedding network:

$$h_{v,m,t} = f_\eta(e_{v,m,t}), \quad (64)$$

where $f_\eta(\cdot)$ denotes the embedding function with trainable parameters η .

To accommodate the variable number of EVs associated with each microgrid, an attention-based pooling mechanism is employed. The attention weight of EV v is calculated as

$$\alpha_{v,m,t} = \frac{\exp(w^T \tanh(W_h h_{v,m,t}))}{\sum_{u \in \mathcal{V}_m^a(t)} \exp(w^T \tanh(W_h h_{u,m,t}))}, \quad (65)$$

and the aggregated EV representation is given by

$$\psi_{m,t}^{EV} = \sum_{v \in \mathcal{V}_m^a(t)} \alpha_{v,m,t} h_{v,m,t}, \quad (66)$$

where W_h and w are trainable parameters. If no EV is associated with microgrid m , $\psi_{m,t}^{EV}$ is set to a zero vector. This attention representation enables the agent to assign higher importance to EVs that are more relevant to local energy management, such as those with high charging demand, severe range-anxiety risk, or imminent departure.

The compact system-level coordination signal received by agent m is defined as

$$s_{m,t}^{sys} = [\bar{P}_{L,-m,t}, \bar{P}_{PV,-m,t}, \bar{P}_{WT,-m,t}, \bar{P}_{EV,-m,t}^{ch}, \bar{P}_{EV,-m,t}^{dis}, \bar{Q}_{-m,t}, \bar{R}_{-m,t}^{risk}, \rho_t^{buy}, \rho_t^{sell}], \quad (67)$$

where $-m$ denotes all microgrids except m . The aggregated load and renewable generation signals are computed as

$$\bar{P}_{\ell,-m,t} = \sum_{j \in \mathcal{M}, j \neq m} P_{\ell,j,t}, \quad \ell \in \{L, PV, WT\}, \quad (68)$$

and the aggregated EV charging and discharging signals are defined as

$$\bar{P}_{EV,-m,t}^{\kappa} = \sum_{j \in \mathcal{M}, j \neq m} P_{EV,j,t}^{\kappa}, \quad \kappa \in \{ch, dis\}. \quad (69)$$

The average queuing condition of the other microgrids is given by

$$\bar{Q}_{-m,t} = \frac{1}{|\mathcal{M}| - 1} \sum_{j \in \mathcal{M}, j \neq m} Q_{j,t}. \quad (70)$$

Similarly, the average range-anxiety risk of EVs associated with the other microgrids is calculated as

$$\bar{R}_{-m,t}^{risk} = \frac{1}{|\mathcal{V}_{-m}^a(t)| + \epsilon} \sum_{j \in \mathcal{M}, j \neq m} \sum_{v \in \mathcal{V}_j^a(t)} R_{v,t}^{risk}, \quad (71)$$

where ϵ is a small positive constant used to avoid division by zero, and

$$\mathcal{V}_{-m}^a(t) = \bigcup_{j \in \mathcal{M}, j \neq m} \mathcal{V}_j^a(t). \quad (72)$$

The coordination signal in Eq. (67) provides each decentralized agent with a compact perception of system-level load pressure, renewable availability, EV charging demand, V2G potential, queue congestion, and mobility-reliability risk, without requiring full global observability.

3.3. Local action space

This study focuses on energy management rather than charging-pile assignment or queue control. Therefore, EV travel, queuing, and physical connection states are treated as exogenous environmental states. Once an EV is physically connected to a microgrid, its charging and discharging powers can be scheduled by the corresponding microgrid agent, subject to the constraints in Eqs. (42) and (43).

The local action of microgrid agent m at interval t is defined as

$$a_{m,t} = [a_{m,t}^{DG}, a_{m,t}^{ESS}, a_{m,t}^{Grid}, a_{m,t}^{DR}, a_{m,t}^{EV}], \quad (73)$$

where the five components correspond to dispatchable DG scheduling, ESS operation, utility-grid interaction, demand response, and EV charging/discharging control, respectively. Specifically,

$$a_{m,t}^{DG} = [\{P_{DG,g,m,t}\}_{g \in G_m}], \quad (74)$$

$$a_{m,t}^{ESS} = [\{P_{b,m,t}^{ch}, P_{b,m,t}^{dis}, u_{b,m,t}^{ch}, u_{b,m,t}^{dis}\}_{b \in B_m}], \quad (75)$$

$$a_{m,t}^{Grid} = [P_{Grid,m,t}^{buy}, P_{Grid,m,t}^{sell}, u_{Grid,m,t}^{buy}, u_{Grid,m,t}^{sell}], \quad (76)$$

$$a_{m,t}^{DR} = [P_{cur,m,t}, P_{shift,m,t}], \quad (77)$$

and

$$a_{m,t}^{EV} = [\{P_{v,m,t}^{ch}, P_{v,m,t}^{dis}\}_{v \in \mathcal{V}_m^c(t)}], \quad (78)$$

where $\mathcal{V}_m^c(t)$ denotes the set of EVs physically connected to microgrid m at interval t , i.e.,

$$\mathcal{V}_m^c(t) = \{v \in \mathcal{V} \mid \delta_{v,m,t}^c = 1\}. \quad (79)$$

Accordingly, the local action space of agent m is hybrid and can be expressed as

$$\mathcal{A}_m = \mathcal{A}_m^d \times \mathcal{A}_m^c, \quad (80)$$

where $\mathcal{A}_m^d$ contains the binary operating-state variables associated with ESS charging/discharging and grid buying/selling, while $\mathcal{A}_m^c$ contains the continuous power-control variables in Eq. (73). The feasible local

action set is determined by the operational constraints of DGs, ESSs, grid transactions, demand response, and EV charging/discharging, as specified in Eqs. (4)–(5), Eqs. (9)–(13), Eqs. (16)–(18), Eqs. (20)–(22), and Eqs. (42)–(49).

The aggregated EV charging and discharging powers entering the microgrid power balance are obtained by summing the individual power decisions of all physically connected EVs:

$$P_{EV,m,t}^{ch} = \sum_{v \in \mathcal{V}_m^c(t)} P_{v,m,t}^{ch}, \quad (81)$$

$$P_{EV,m,t}^{dis} = \sum_{v \in \mathcal{V}_m^c(t)} P_{v,m,t}^{dis}. \quad (82)$$

These aggregated variables serve as the interface between the local EV charging/discharging actions and the microgrid-side power balance constraint.

3.4. State transition

Let the joint action of all microgrid agents at scheduling interval t be denoted by

$$\mathbf{a}_t = [a_{1,t}, a_{2,t}, \dots, a_{|\mathcal{M}|,t}]. \quad (83)$$

Given the current global state s_t and the joint action $\mathbf{a}_t$, the system evolves according to the transition kernel

$$s_{t+1} \sim \mathcal{P}(s_{t+1} \mid s_t, \mathbf{a}_t). \quad (84)$$

The transition process captures both controllable dynamics, induced by dispatching actions, and exogenous uncertainties, including renewable generation, load variation, electricity prices, traffic conditions, and EV arrival behavior.

The SOC transitions of ESSs and EVs are governed by Eqs. (12) and (47), respectively. The EV mobility states evolve according to the transportation process, arrival behavior, queuing status, and physical connection status, while satisfying the mutually exclusive state condition in Eq. (28). In this formulation, $\delta_{v,m,t}^q$ and $\delta_{v,m,t}^c$ are not direct control actions of microgrid agents; instead, they are updated by the environment according to the mobility and charging-access processes.

For reward evaluation and feasibility assessment, the power-balance residual of microgrid m is defined as

$$\begin{aligned} \Delta P_{m,t}^{bal} = & \sum_{g \in G_m} P_{DG,g,m,t} + \sum_{b \in B_m} (P_{b,m,t}^{dis} - P_{b,m,t}^{ch}) + P_{PV,m,t} \\ & + P_{WT,m,t} + P_{Grid,m,t}^{buy} - P_{Grid,m,t}^{sell} \\ & + P_{EV,m,t}^{dis} - P_{EV,m,t}^{ch} - P_{L,m,t}. \end{aligned} \quad (85)$$

Accordingly, the microgrid power-balance constraint in Eq. (26) is satisfied when

$$\Delta P_{m,t}^{bal} = 0, \quad \forall m \in \mathcal{M}, t \in \mathcal{T}. \quad (86)$$

3.5. Reward function

The cooperative reward is designed to align the learning objective with the original optimization objective in Eq. (1). The stage-wise microgrid-side cost is defined according to Eq. (2) as

$$C_t^{MG} = \sum_{m \in \mathcal{M}} (C_{DG,m,t} + C_{Grid,m,t} + C_{ESS,m,t}^{deg} + C_{DR,m,t} + C_{CO_2,m,t}), \quad (87)$$

and the stage-wise EV-side cost is given by

$$C_t^{EV} = C_{EV,t}^{mob} + C_{EV,t}^{deg} + C_{EV,t}^{risk}. \quad (88)$$

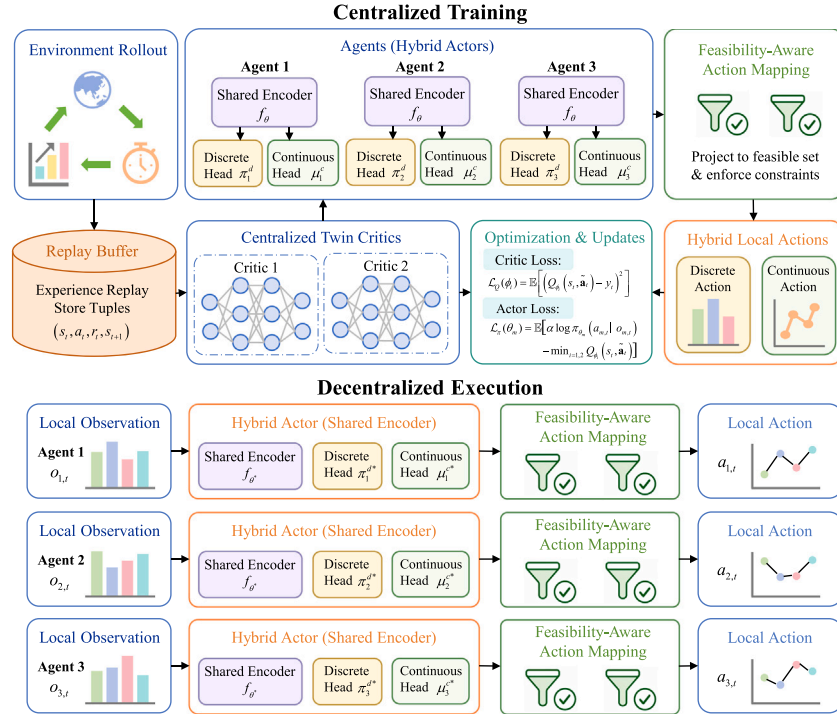


Fig. 3. Overall architecture of the CTDE-based multi-agent Deep reinforcement learning algorithm. During centralized training, hybrid actors generate raw local actions, which are projected onto the feasible action set through the feasibility-aware action mapping layer, while centralized twin critics are updated using replayed transitions. During decentralized execution, each microgrid agent determines its local action independently based on its local observation using the learned hybrid actor and the same feasibility-aware mapping mechanism.

Accordingly, the cooperative reward at interval t is formulated as

$$r_t = -(\omega_{MG} C_t^{MG} + \omega_{EV} C_t^{EV}) - \Phi_t, \quad (89)$$

where Φ_t denotes the penalty imposed on constraint violations. Specifically, it is defined as

$$\Phi_t = \lambda_{pb} \sum_{m \in \mathcal{M}} |\Delta P_{m,t}^{bal}| + \lambda_{op} \Phi_t^{op} + \lambda_{EV} \Phi_t^{EV}, \quad (90)$$

where the first term penalizes the violation of the microgrid power-balance constraint, and Φ_t^{op} and Φ_t^{EV} represent the penalty terms associated with operational constraints and EV-related constraints, respectively. The coefficients λ_{pb} , λ_{op} , and λ_{EV} are nonnegative penalty weights.

The learning objective is to determine the joint decentralized policy $\Pi = \{\pi_{\theta_m}\}_{m \in \mathcal{M}}$ that maximizes the expected discounted cooperative return, i.e.,

$$\Pi^* = \arg \max_{\Pi} \mathbb{E}_{\Pi} \left[\sum_{t=1}^T \gamma^{t-1} r_t \right], \quad (91)$$

where the expectation is taken over the trajectories induced by the joint policy Π and the transition kernel in Eq. (84). This reward design converts the original cost-minimization problem into an equivalent cooperative return-maximization problem while explicitly accounting for constraint feasibility.

The proposed multi-agent MDP reformulation converts the original deterministic optimization model into a scalable sequential decision-making framework. Each microgrid agent dispatches its local controllable resources and the charging/discharging powers of physically connected EVs, whereas EV mobility, queueing, and connection states are updated by the environment. By integrating attention-based EV aggregation with compact inter-microgrid coordination signals, the formulation

supports decentralized execution while preserving essential system-level coupling information. This provides the decision-making basis for the customized MA-DRL algorithm presented in the next section.

4. Multi-agent DRL algorithm

To address the cooperative scheduling problem of MMG energy management with mobile EV participation, a customized HEC-MADRL algorithm is developed under the centralized-training and decentralized-execution (CTDE) paradigm, as shown in Fig. 3. During training, centralized critics utilize system-level information and the joint actions of all microgrid agents to evaluate the cooperative return. During execution, each microgrid agent independently generates its local scheduling action based only on its local observation defined in Eq. (61). This structure is consistent with practical MMG operation, where each microgrid generally has access to local measurements and limited coordination information rather than complete system-wide data.

The proposed algorithm is specifically designed for the coupled microgrid-EV energy management task. The decentralized actors take the EV-aware local observation as input, where the attention-based aggregation in Eq. (66) enables each agent to represent a variable number of EVs associated with the corresponding microgrid. Moreover, the policy structure is designed to accommodate the hybrid action space in Eq. (80), which consists of binary operating states and continuous dispatching variables. To further enhance the physical consistency of the generated scheduling decisions, a feasibility-aware action mapping layer is incorporated into the decision-making process.

These components form a sequential and jointly optimized decision pipeline. The EV-aware attention module first extracts mobility- and availability-dependent EV information and embeds it into the local observation. The hybrid actor then converts this representation into discrete operating modes and continuous power commands. The resulting actions are corrected by the feasibility- and risk-aware enhancement module before execution, after which the corrected joint

actions are evaluated by the centralized twin critics. The critic feedback is propagated through the actor to the attention module during training, thereby coupling EV representation, hybrid decision generation, constraint handling, and cooperative value learning within a unified framework.

4.1. Hybrid decentralized actor

For each microgrid agent m , the decentralized actor maps the local observation $o_{m,t}$ to a hybrid scheduling action. To accommodate the mixed discrete–continuous structure in Eq. (80), the policy is factorized into a discrete operating-mode policy and a mode-conditioned continuous power-dispatch policy:

$$\pi_{\theta_m}(a_{m,t} | o_{m,t}) = \pi_{\theta_m}^d(a_{m,t}^d | o_{m,t}) \pi_{\theta_m}^c(a_{m,t}^c | o_{m,t}, a_{m,t}^d), \quad (92)$$

where $\pi_{\theta_m}^d(\cdot)$ and $\pi_{\theta_m}^c(\cdot)$ denote the discrete and continuous policy heads, respectively, and θ_m represents the actor-network parameters. The discrete head first determines the operating modes of fixed stationary devices, after which the continuous head generates the corresponding power magnitudes and other continuously adjustable commands.

The hybrid structure is applied selectively according to the physical role of each resource. For ESSs and utility-grid interaction, charging/discharging and buying/selling are fixed-dimensional, mutually exclusive supervisory modes that are explicitly represented in the original scheduling formulation. Retaining these modes as discrete actions preserves their operational interpretation and allows the continuous power bounds to be conditioned on the selected mode before dispatch. Compared with determining the operating direction solely from the sign of a continuous action, this mode–magnitude decomposition avoids ambiguous bidirectional commands, reduces switching around the zero-power boundary during exploration, and decreases reliance on post-processing corrections.

The discrete policy head generates the binary operating variables associated with ESS charging/discharging and grid buying/selling. Let $z_{n,m,t}$ denote the n -th hard discrete action variable of agent m during interval t , and let C_n denote its candidate set. For binary decisions, $C_n = \{0, 1\}$. During training, a straight-through Gumbel–Softmax estimator is adopted to obtain a differentiable relaxation of $z_{n,m,t}$. Specifically, the relaxed probability of selecting candidate $r \in C_n$ is given by

$$\tilde{z}_{n,r,m,t} = \frac{\exp[(\ell_{n,r,m,t} + g_{n,r})/\tau_g]}{\sum_{r' \in C_n} \exp[(\ell_{n,r',m,t} + g_{n,r'})/\tau_g]}, \quad r \in C_n, \quad (93)$$

where $\ell_{n,r,m,t}$ denotes the logit associated with candidate r , $g_{n,r}$ is a Gumbel noise sample, and τ_g is the temperature parameter. Here, $\tilde{z}_{n,r,m,t}$ is the differentiable representation used for gradient backpropagation, whereas $z_{n,m,t}$ is the hard action used in the forward decision process:

$$z_{n,m,t} = \arg \max_{r \in C_n} \tilde{z}_{n,r,m,t}. \quad (94)$$

During decentralized execution, the Gumbel noise is removed, and the action with the highest predicted probability is selected deterministically.

EV actions are represented differently because traveling, queuing, and physical connection are environment-driven mobility states rather than operating modes selected by the agent. Moreover, the number of physically connected EVs varies over time. Assigning a discrete charging/idle/discharging mode to every connected EV would therefore introduce a time-varying and combinatorial discrete action space. Accordingly, each connected EV is controlled through an individual signed continuous net-power action,

$$p_{v,m,t}^{EV} = p_{v,m,t}^{dis} - p_{v,m,t}^{ch}, \quad (95)$$

where $p_{v,m,t}^{EV} > 0$, $p_{v,m,t}^{EV} < 0$, and $p_{v,m,t}^{EV} = 0$ denote discharging, charging, and idling, respectively. Thus, charging/discharging mutual exclusivity

is determined directly by the sign of the action, while the connection-state mask forces the power to zero for traveling or queuing EVs. The actor generates individual EV commands, which are aggregated only when constructing the microgrid power balance.

The continuous policy head therefore generates DG outputs, mode-conditioned ESS and grid power magnitudes, demand-response quantities, and individual EV net powers. A squashed Gaussian policy is adopted to ensure that these continuous actions remain within their admissible intervals:

$$a_{m,t}^c = \bar{a}_{m,t}^c + \Delta a_{m,t}^c \odot \tanh(\xi_{m,t}), \quad (96)$$

$$\xi_{m,t} \sim \mathcal{N}\left(\mu_{\theta_m}(o_{m,t}, a_{m,t}^d), \text{diag}\left[\sigma_{\theta_m}^2(o_{m,t}, a_{m,t}^d)\right]\right).$$

Here, $\mu_{\theta_m}(\cdot)$ and $\sigma_{\theta_m}(\cdot)$ are the mean and standard-deviation outputs of the continuous policy head, respectively, and $\odot$ denotes element-wise multiplication. The vectors $\bar{a}_{m,t}^c$ and $\Delta a_{m,t}^c$ denote the center and half-width of the state- and mode-dependent action interval:

$$\bar{a}_{m,t}^c = \frac{\bar{a}_{m,t}^c + a_{m,t}^c}{2}, \quad \Delta a_{m,t}^c = \frac{\bar{a}_{m,t}^c - a_{m,t}^c}{2}. \quad (97)$$

The bounds are dynamically derived from the operational constraints of DGs, ESSs, grid interaction, demand response, and EV charging/discharging. For ESS and grid powers, the intervals depend on the operating modes selected by the discrete head. For EV actions, the bounds are jointly determined by the connection state, power rating, and current SOC. Consequently, discrete actions are used only for the fixed supervisory modes of ESSs and grid interaction, whereas continuous actions are used for power magnitudes and dynamically varying EV dispatch.

4.2. Centralized twin-critic learning

To enhance value-estimation stability in the high-dimensional hybrid joint action space, a centralized twin-critic structure is adopted. Due to function approximation errors, stochastic EV mobility, simultaneous policy updates among agents, and the presence of physical operating constraints, a single critic may lead to overestimated action values. Therefore, two centralized critics are introduced as $Q_{\phi_i}(s_t, \mathbf{a}_t)$, $i = 1, 2$, where ϕ_i denotes the parameters of the i -th critic. Each critic takes the global state s_t and the joint action $\mathbf{a}_t$ as inputs and approximates the expected cooperative return:

$$Q_{\phi_i}(s_t, \mathbf{a}_t) \approx \mathbb{E}_{\Pi} \left[\sum_{\tau=t}^T \gamma^{\tau-t} r_{\tau} \mid s_t, \mathbf{a}_t \right], \quad i = 1, 2. \quad (98)$$

For a transition sampled from the replay buffer $\mathcal{D}$, the next-step joint action is generated by the target actors and subsequently projected by the feasibility-aware action mapping layer. The target value is then computed using the clipped double-Q mechanism:

$$y_t = r_t + \gamma(1 - d_t) \left[\min_{i=1,2} Q_{\phi_i}(s_{t+1}, \tilde{\mathbf{a}}_{t+1}) - \alpha \sum_{m \in \mathcal{M}} \log \pi_{\theta_m}(\tilde{a}_{m,t+1} | o_{m,t+1}) \right], \quad (99)$$

where Q_{ϕ_i} denotes the i -th target critic, π_{θ_m} is the target actor of agent m , $\tilde{\mathbf{a}}_{t+1}$ is the feasible joint action after action mapping, α is the entropy temperature coefficient, and d_t is the terminal indicator. The minimum operator in Eq. (99) provides a conservative target estimate and mitigates overestimation bias in centralized value learning.

For the hybrid policy, the log-probability term in Eq. (99) is evaluated as the sum of the discrete and continuous policy components:

$$\log \pi_{\theta_m}(\tilde{a}_{m,t+1} | o_{m,t+1}) = \log \pi_{\theta_m}^d(\tilde{a}_{m,t+1}^d | o_{m,t+1}) + \log \pi_{\theta_m}^c(\tilde{a}_{m,t+1}^c | o_{m,t+1}, \tilde{a}_{m,t+1}^d). \quad (100)$$

This treatment is consistent with the factorized hybrid actor in Eq. (92).

The critic networks are updated by minimizing the temporal-difference loss:

$$\mathcal{L}_Q(\phi_i) = \mathbb{E}_{(s_t, \tilde{\mathbf{a}}_t, r_t, s_{t+1}, d_t) \sim D} \left[\left(Q_{\phi_i}(s_t, \tilde{\mathbf{a}}_t) - y_t \right)^2 \right], i = 1, 2. \quad (101)$$

Here, $\tilde{\mathbf{a}}_t$ denotes the feasible joint action stored in the replay buffer after the feasibility-aware action mapping. This update enables the centralized critics to evaluate the coupled effects of all microgrid agents while reducing overestimation bias through the twin-critic target.

4.3. Decentralized actor update

Each decentralized actor is updated with guidance from the centralized twin critics, while the execution policy remains fully decentralized. For agent m , the actor first samples a hybrid local action from its factorized policy and then obtains a feasible action $\tilde{a}_{m,t}$ through the feasibility-aware action mapping layer. When updating θ_m , the actions of the other agents are sampled from their current policies and treated as fixed with respect to θ_m . The actor loss is formulated as

$$\mathcal{L}_\pi(\theta_m) = \mathbb{E}_{s_t \sim D, \mathbf{a}_t \sim \Pi} \left[\alpha \log \pi_{\theta_m}(a_{m,t} | o_{m,t}) - \min_{i=1,2} Q_{\phi_i}(s_t, \tilde{\mathbf{a}}_t) \right], \quad (102)$$

where $\tilde{\mathbf{a}}_t = [\tilde{a}_{1,t}, \tilde{a}_{2,t}, \dots, \tilde{a}_{|\mathcal{M}|,t}]$ denotes the feasible joint action after action mapping, and α is the entropy temperature coefficient. The log-probability term is evaluated according to the factorized hybrid policy in Eq. (100). The entropy term encourages exploration, while the centralized critics provide system-level value guidance for cooperative scheduling.

After training, the centralized critics are discarded. During decentralized execution, each microgrid agent generates its local scheduling action solely from its local observation $o_{m,t}$, which is followed by the feasibility-aware action mapping layer to satisfy the physical operating constraints.

The target networks are updated by soft replacement:

$$\bar{\phi}_i \leftarrow \tau_s \phi_i + (1 - \tau_s) \bar{\phi}_i, \quad i = 1, 2, \quad (103)$$

$$\bar{\theta}_m \leftarrow \tau_s \theta_m + (1 - \tau_s) \bar{\theta}_m, \quad m \in \mathcal{M}, \quad (104)$$

where $\tau_s \in (0, 1)$ denotes the soft-update coefficient. This soft-update mechanism stabilizes target-value estimation by slowly tracking the online critics and actors.

4.4. Feasibility- and risk-aware action enhancement

The actions generated by neural networks may violate physical operating constraints, especially in the presence of hybrid decision variables, intertemporal SOC dynamics, and time-varying EV connection states. To improve the physical consistency of the learned policy, a feasibility-aware action mapping layer is applied after the actor's output:

$$\tilde{a}_{m,t} = F_m(a_{m,t}^{raw}, o_{m,t}), \quad (105)$$

where $a_{m,t}^{raw}$ is the raw action generated by the actor, $\tilde{a}_{m,t}$ is the feasible action executed by agent m , and $F_m(\cdot)$ denotes the mapping operator.

The mapping layer does not introduce additional decision variables. Instead, it projects or masks the raw action according to the operational constraints defined in the problem formulation. Specifically, it enforces the DG output and ramping constraints in Eqs. (4) and (5), the ESS operating constraints in Eqs. (9)–(13), the grid-interaction constraints in Eqs. (16)–(18), the demand-response constraints in Eqs. (20)–(22), and the EV-related constraints in Eqs. (42)–(49). For EV charging/discharging actions, the mapping layer further applies the connection-state mask in Eq. (79), such that only physically connected EVs can receive nonzero charging or discharging commands.

The power-balance residual in Eq. (85) is also evaluated after action mapping. When the residual cannot be fully eliminated due to local operating limits, the remaining infeasibility is penalized through the constraint-violation penalty in Eq. (90). This design combines hard correction for simple device-level constraints with soft penalization for residual coupled infeasibility.

Although the EV-side range-anxiety cost is included in the co-operative reward, excessive V2G discharging may still occur during early-stage exploration. To discourage aggressive discharging of EVs with high mobility-reliability risk, an auxiliary EV-aware regularization term is incorporated into the actor objective:

$$\mathcal{L}_\pi^{reg}(\theta_m) = \mathcal{L}_\pi(\theta_m) + \lambda_{RA} \mathbb{E}_{s_t \sim D} \left[\sum_{v \in \mathcal{V}_m^C(t)} R_{v,t}^{risk} \tilde{P}_{v,m,t}^{dis} \Delta t \right], \quad (106)$$

where λ_{RA} is the regularization coefficient, and $\tilde{P}_{v,m,t}^{dis}$ denotes the feasible EV discharging power after action mapping. Since $R_{v,t}^{risk}$ is defined in Eq. (52), this regularization term is consistent with the EV-side range-anxiety model and does not introduce additional behavioral assumptions.

4.5. Training procedure

The proposed algorithm is trained using off-policy experience replay. At the beginning of each episode, the microgrid-EV environment is initialized. At each scheduling interval, each microgrid agent constructs its local observation according to Eq. (61), generates a raw hybrid action through its decentralized actor, and obtains a feasible action via the mapping layer in Eq. (105). The joint feasible action is then executed in the environment. Subsequently, the system state evolves according to Eq. (84), and the cooperative reward is calculated using Eq. (89).

The transition tuple $(s_t, \tilde{\mathbf{a}}_t, r_t, s_{t+1}, d_t)$ is stored in the replay buffer D , where $\tilde{\mathbf{a}}_t$ denotes the feasible joint action and d_t is the terminal indicator. During training, mini-batches are sampled from D to update the centralized twin critics using Eq. (101). The decentralized actors are then updated using the regularized actor objective in Eq. (106). Finally, the target critics and target actors are updated through the soft-update rules in Eqs. (103) and (104). The overall procedure is summarized in Algorithm 1.

The proposed HEC-MADRL algorithm exploits the structural characteristics of the microgrid-EV scheduling problem. Specifically, the attention-based EV representation enables each agent to handle a variable number of associated EVs, and the hybrid actor accommodates both binary operating states and continuous dispatching variables without full action-space discretization. The centralized twin critics stabilize cooperative value learning under inter-microgrid coupling, whereas the feasibility-aware mapping and EV-aware regularization improve physical consistency and mitigate excessive V2G discharging for high-risk EVs. Consequently, the proposed method provides a scalable and constraint-aware decision-making framework for real-time MMG energy management with mobile EV resources.

5. Case studies

5.1. Experimental setup

The proposed method is evaluated on a residential MMG system with mobile EV participation. The base system contains three microgrids, each equipped with dispatchable DGs, PV and WT units, ESSs, demand response resources, grid-interaction interfaces, and EV charging/discharging facilities. A one-day scheduling horizon is discretized into $T = 96$ intervals with $\Delta t = 15$ min, and each training episode represents one daily operation cycle.

The experiments are conducted using residential profiles from the U.S. Pecan Street database [33,34]. The training set covers February 1, 2018 to July 31, 2018, and the testing set covers August 1, 2018 to September 28, 2018. All time-series data are aligned to the 15-min

Algorithm 1: Customized HEC-MADRL algorithm.

Input: Microgrid–EV environment; replay buffer D ; discount factor γ ; entropy temperature α ; soft-update coefficient τ_s ; mini-batch size N_b ; number of episodes N_{ep} .

Output: Decentralized actor policies $\{\pi_{\theta_m}\}_{m \in \mathcal{M}}$.

- 1 Initialize actors $\{\pi_{\theta_m}\}$, centralized critics Q_{ϕ_1} and Q_{ϕ_2} , and their target networks $\{\pi_{\bar{\theta}_m}\}$, $Q_{\bar{\phi}_1}$, and $Q_{\bar{\phi}_2}$;
- 2 Set $\bar{\theta}_m \leftarrow \theta_m, \forall m \in \mathcal{M}$, and $\bar{\phi}_i \leftarrow \phi_i, i = 1, 2$;
- 3 Initialize replay buffer D ;
- 4 **for** $episode = 1, \dots, N_{ep}$ **do**
- 5 Initialize the microgrid–EV environment and obtain s_1 ;
- 6 **for** $t = 1, \dots, T$ **do**
- 7 **foreach** microgrid agent $m \in \mathcal{M}$ **do**
- 8 Construct local observation $o_{m,t}$ using Eq. (61);
- 9 Generate a raw hybrid action $a_{m,t}^{raw}$ using the decentralized actor;
- 10 Map $a_{m,t}^{raw}$ into a feasible action $\tilde{a}_{m,t}$ using Eq. (105);
- 11 **end**
- 12 Execute the feasible joint action $\tilde{\mathbf{a}}_t = [\tilde{a}_{1,t}, \dots, \tilde{a}_{|\mathcal{M}|,t}]$;
- 13 Observe r_t, s_{t+1} , and terminal indicator d_t ;
- 14 Store $(s_t, \tilde{\mathbf{a}}_t, r_t, s_{t+1}, d_t)$ in D ;
- 15 **if** $|D| \geq N_b$ **then**
- 16 Sample a mini-batch from D ;
- 17 Generate feasible next actions using target actors and the action mapping layer;
- 18 Compute the target value y_t using Eq. (99);
- 19 Update centralized critics by minimizing Eq. (101);
- 20 Update decentralized actors by minimizing Eq. (106);
- 21 Softly update target critics and target actors using Eqs. (103) and (104);
- 22 **end**
- 23 **end**
- 24 **end**
- 25 **return** $\{\pi_{\theta_m}\}_{m \in \mathcal{M}}$;

Table 2

Base MMG system parameters.

Parameter	MG1	MG2	MG3
Peak load (kW)	92	86	78
DG rated power (kW)	65	60	55
DG minimum output ratio	0.20	0.20	0.20
PV installed capacity (kW)	42	38	35
WT rated power (kW)	24	22	20
ESS capacity (kWh)	150	130	120
ESS power limit (kW)	45	40	36
ESS charging/discharging efficiency	0.95/0.95	0.95/0.95	0.95/0.95
ESS SOC range	0.10–0.90	0.10–0.90	0.10–0.90
Number of EVs	20	18	16
EV battery capacity (kWh)	50	50	50
EV charging limit (kW)	7.2	7.2	7.2
EV discharging limit (kW)	6.6	6.6	6.6
EV charging/discharging efficiency	0.95/0.95	0.95/0.95	0.95/0.95
EV SOC range	0.15–0.95	0.15–0.95	0.15–0.95
Required departure SOC	0.75	0.75	0.75

resolution. Missing values shorter than one hour are filled by linear interpolation, whereas days with long missing segments are discarded. Continuous inputs are normalized to $[0, 1]$ for neural-network training, while physical values are retained for action execution, cost evaluation, and feasibility checking. The base-system parameters are summarized in Table 2. For scalability analysis, the base system is extended to 5, 10, and 20 microgrids by resampling residential load, PV profiles, and EV mobility patterns from the same statistical structure.

Table 3

Time-of-use electricity prices.

Time	08:00–14:00	14:00–20:00	20:00–22:00	22:00–08:00
Buying price (\$/kWh)	0.28	0.48	0.28	0.12
Selling price (\$/kWh)	0.14	0.24	0.14	0.06

Table 4

EV mobility profile settings.

Item	Setting
Arrival time distribution	17:00–22:00
Departure time distribution	06:00–09:00
Desired arrival time deviation	0–30 min
Initial SOC range	0.25–0.65
Required departure SOC	0.75
Expected trip distance	20–80 km
EV battery capacity	50 kWh
Maximum charging power	7.2 kW
Maximum discharging power	6.6 kW
Minimum SOC	0.15
Maximum SOC	0.95
Connected-EV control	Enabled only when $\delta_{i,m,t}^c = 1$

Table 5

Demand response and carbon-related parameters.

Parameter	Value
Maximum load curtailment ratio μ_{cur}	0.10
Maximum load shifting ratio μ_{shift}	0.15
Curtailment compensation coefficient c_{cur}	0.35 \$/kWh
Load shifting compensation coefficient c_{shift}	0.08 \$/kWh
Carbon price ρ_{CO_2}	0.06 \$/kg
DG carbon emission factor	0.62 kg/kWh
Grid carbon emission factor	0.48 kg/kWh

As shown in Table 3, a time-of-use tariff is adopted to evaluate the response of different methods to time-varying electricity prices. The selling price is set lower than the buying price in each period to avoid trivial arbitrage. EV mobility profiles are generated according to typical residential travel patterns, and the corresponding settings are reported in Table 4. The demand response and carbon-related parameters are given in Table 5. These settings provide moderate load-side flexibility and enable the evaluation of low-carbon scheduling.

5.2. Implementation details and evaluation protocol

All algorithms are implemented in Python 3.7.0 with PyTorch 1.2 and executed on a workstation equipped with a Core Ultra 9 285 K CPU and an NVIDIA GeForce RTX 5090 GPU. For the proposed method, each microgrid actor is implemented as an attention-enhanced hybrid MLP. The EV features are first aggregated using Eq. (66), concatenated with the remaining local observations, and then processed by four fully connected hidden layers with 128 ReLU neurons. The discrete and continuous policy heads follow Eqs. (93) and (96), respectively. Each centralized critic is implemented as a four-layer MLP with 128 ReLU neurons and maps the global state–joint action pair to a scalar state–action value. The main training hyperparameters are listed in Table 6.

The proposed method is compared with representative model-based and learning-based baselines, including MPC, MADDPG, independent DRL, centralized MASAC, VA2 C, and MAAC-PER. MPC [35] is adopted as a rolling-horizon model-based optimization benchmark due to its wide use in energy management problems. MADDPG [36] is selected to represent deterministic multi-agent policy learning under centralized training and decentralized execution. Independent DRL [5] trains each microgrid agent separately without centralized value estimation, thereby evaluating the effect of ignoring inter-agent coupling. Centralized MASAC [37] extends the soft actor–critic framework to

Table 6
Training hyperparameter settings.

Hyperparameter	Value
Critic learning rate	3×10^{-3}
Actor learning rate	1.5×10^{-4}
Discount factor γ	0.99
Soft update coefficient τ_s	5×10^{-4}
Optimizer	Adam
Mini-batch size	256
Replay buffer capacity	1×10^6
Number of hidden layers	4
Hidden units per layer	128
Activation function	ReLU
Network update frequency	10
Gumbel-Softmax temperature τ_g	0.5
Entropy temperature coefficient α	0.2
EV-aware regularization coefficient λ_{RA}	0.08
Training episodes	3000
Number of random seeds	5

multi-agent cooperative learning and is used to assess the benefit of entropy-regularized stochastic policies. MAAC-PER [38] represents an attention-based multi-agent critic method with prioritized experience replay. For a fair comparison, all learning-based baselines use the same state, action, and reward definitions whenever applicable, and are trained and tested on the same datasets without further parameter tuning during evaluation.

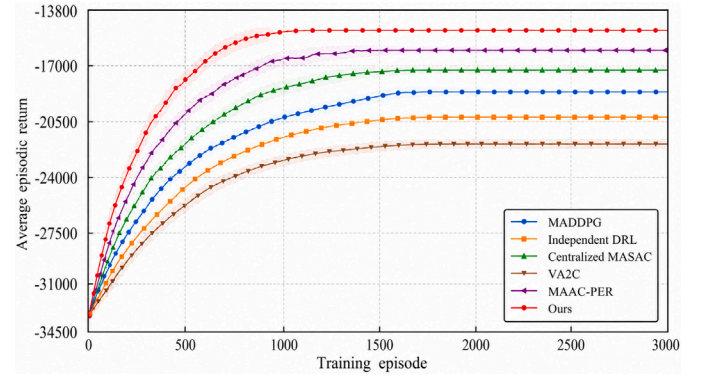
For evaluation, all methods are tested on the same testing set without further parameter updates. The performance metrics used in the following tables include the total objective value J , microgrid-side cost J_{MG} , EV-side cost J_{EV} , total carbon emissions $E_{CO_2}^{total}$, accumulated power-balance violation V_{bal} , departure-SOC satisfaction rate η_{dep} , average range-anxiety risk $\bar{R}^{risk}$, renewable-energy utilization rate, grid import/export energies, online decision time T_{dec} , training time, memory usage, and convergence episodes. Specifically, J , J_{MG} , J_{EV} , and $E_{CO_2}^{total}$ are accumulated over the entire testing horizon. The metric V_{bal} is calculated by accumulating the absolute power-balance residual over all microgrids and scheduling intervals. The departure-SOC satisfaction rate η_{dep} denotes the percentage of EVs whose SOC is not lower than the required departure SOC at their departure times, while $\bar{R}^{risk}$ represents the average range-anxiety risk over all EVs and scheduling intervals. The renewable-energy utilization rate is defined as the ratio of consumed renewable energy to available renewable generation over the testing horizon. Grid import and export energies are accumulated over the testing horizon, and T_{dec} denotes the average online decision time required to generate one scheduling action for all microgrid agents in one interval. For convergence analysis, the episodic return is averaged over five independent random seeds.

5.3. Experimental results and analysis

5.3.1. Convergence and training efficiency

Fig. 4 and Table 7 compare the convergence behavior and training efficiency of different learning-based methods. As shown in Fig. 4, all methods improve their episodic returns during training, but the proposed method reaches a stable and higher-return region more rapidly. Quantitatively, our proposed approach converges within 980 episodes, reducing the required number of episodes for convergence by 40.6%, 29.0%, 35.1%, 46.4%, and 24.0% compared with MADDPG, Independent DRL, Centralized MASAC, VA2 C, and MAAC-PER, respectively.

The training-time results in Table 7 further demonstrate the computational advantage of the proposed method. Ours requires 3.54 h for training, which is 14.1%, 3.8%, 18.8%, 22.7%, and 11.9% shorter than MADDPG, Independent DRL, Centralized MASAC, VA2 C, and MAAC-PER, respectively. Although its memory usage is 14.18 GB, it remains comparable to the baselines and is lower than those of MADDPG,

**Fig. 4.** Training convergence curves for different learning-based methods.**Table 7**
Training performance comparison.

Method	Time (h)	Memory (GB)	Conv. Ep.
MADDPG	4.12	14.31	1650
Independent DRL	3.68	15.62	1380
Centralized MASAC	4.36	13.89	1510
VA2 C	4.58	12.74	1830
MAAC-PER	4.02	14.76	1290
Ours	3.54	14.18	980

Independent DRL, and MAAC-PER. These results indicate that the proposed method improves convergence speed and training efficiency without introducing excessive memory overhead.

Overall, the faster convergence and lower training time verify the effectiveness of the proposed multi-agent learning framework. The centralized training mechanism enables more informative policy updates by capturing the coupling among microgrids, EVs, ESSs, renewable generation, and grid transactions, thereby improving both convergence stability and learning efficiency.

5.3.2. Overall testing performance

Table 8 reports the overall testing performance of all compared methods. The proposed MA-DRL method achieves the best values in terms of total objective value, microgrid-side cost, EV-side cost, carbon emissions, power-balance violation, departure-SOC satisfaction rate, and range-anxiety risk. Compared with MAAC-PER, the strongest baseline, the proposed method reduces J from \$15,060.4 to \$14,392.7, corresponding to a 4.43% reduction. It also decreases $E_{CO_2}^{total}$ by 5.13%, reduces V_{bal} by 74.47%, and lowers $\bar{R}^{risk}$ by 39.47%. Meanwhile, the departure-SOC satisfaction rate is improved from 97.7% to 99.0%, indicating enhanced EV mobility reliability.

The MPC benchmark provides competitive scheduling quality, but its online decision time reaches 38.72 s, which is unsuitable for real-time implementation compared with the proposed method that requires only 0.043 s. Among learning-based baselines, MADDPG shows the worst objective value due to unstable value estimation in the hybrid joint action space. Independent DRL achieves low decision latency but lacks coordinated value learning among microgrids. Centralized MASAC improves inter-agent coordination, whereas its centralized action representation increases online computational burden. MAAC-PER performs better than most baselines by introducing attention-based critic learning and prioritized experience replay. In contrast, the proposed method further improves economic performance, low-carbon operation, constraint satisfaction, and EV-side reliability by integrating EV-aware representation, feasibility-aware action mapping, and centralized twin-critic learning.

Table 8
Overall performance comparison on the testing set.

Method	J (\$)	J_{MG} (\$)	J_{EV} (\$)	$E_{CO_2}^{total}$ (kg)	V_{bal} (kWh)	η_{dep} (%)	$\bar{R}^{risk}$	T_{dec} (s)
MPC	15,187.4	11,320.6	3866.8	856.4	3.9	97.2	0.035	38.72
MADDPG	16,480.9	11,990.3	4490.6	925.7	12.6	93.5	0.069	0.056
Independent DRL	15,892.1	11,632.8	4259.3	890.1	9.8	94.8	0.059	0.041
Centralized MASAC	15,366.5	11,260.9	4105.6	862.5	6.1	96.5	0.046	0.074
VA2 C	15,935.8	11,685.4	4250.4	899.2	8.7	95.1	0.057	0.062
MAAC-PER	15,060.4	11,102.7	3957.7	839.6	4.7	97.7	0.038	0.067
Ours	14,392.7	10,610.5	3782.2	796.5	1.2	99.0	0.023	0.043

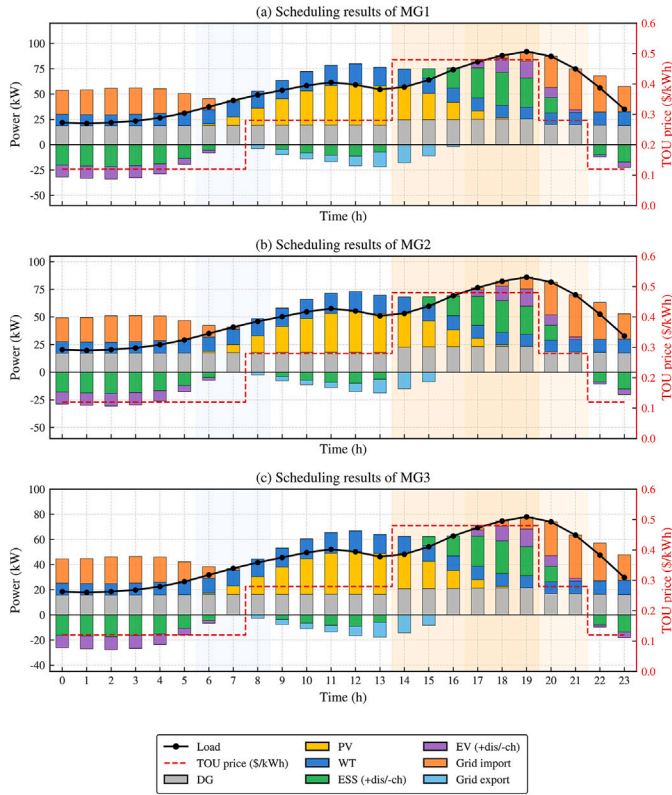


Fig. 5. Representative scheduling results of three microgrids on a test day. (a) scheduling results of MG1. (b) scheduling results of MG2. (c) scheduling results of MG3. The positive and negative power components denote power supply and absorption, respectively, while the TOU electricity price and peak-price periods are also indicated.

5.3.3. Representative-day scheduling results

Fig. 5 presents the representative daily scheduling results of MG1–MG3. The scheduled power components satisfy the hourly power-balance constraint, where DG, PV, WT, ESS/EV discharging, and grid import act as positive power components, while ESS/EV charging and grid export act as negative components. The three microgrids are parameter heterogeneous, with peak loads of 92, 86, and 78 kW for MG1, MG2, and MG3, respectively, as well as different generation capacities, ESS capacities, and EV fleet sizes. These differences create distinct operating ranges and local flexibility portfolios. Meanwhile, the microgrids exhibit similar temporal responses because they are exposed to the same TOU tariff and share the same general operational rules. The different dispatch magnitudes therefore demonstrate that the learned policy can adapt a common scheduling logic to heterogeneous capacity configurations.

The obtained schedules show clear responses to the TOU price signal. During the off-peak period of 22:00–08:00, ESSs and connected EVs

are mainly charged to store low-cost electricity and prepare for subsequent high-price intervals. During renewable-rich daytime hours, PV and WT generation are preferentially utilized to supply local demand, while ESSs absorb part of the surplus renewable energy when local generation exceeds the instantaneous load. In the peak-price period of 14:00–20:00, ESSs and available EVs are discharged to support local demand and reduce expensive grid imports. This charging–discharging pattern demonstrates that the proposed policy performs intertemporal energy shifting rather than myopic power balancing.

It is also observed that grid export is nearly eliminated during 16:00–21:00. This result is consistent with the practical operating condition that the evening peak load, EV arrival, and high electricity price occur simultaneously. Instead of selling energy back to the grid, the proposed policy prioritizes local consumption of renewable generation and stored energy to reduce grid dependence. Grid transactions are therefore mainly used to compensate for residual power mismatch, rather than serving as the primary energy source or sink.

The EV dispatch behavior further demonstrates the importance of incorporating mobility constraints into the decision process. Because EV arrivals occur during 17:00–22:00 and departures occur during 06:00–09:00, charging and discharging actions are activated only when the vehicles are physically connected to the corresponding microgrids. This prevents infeasible EV scheduling during traveling and queuing periods and enables mobile storage flexibility to be utilized without compromising mobility requirements. Overall, the results show that the proposed MA-DRL policy generates feasible, price-responsive, and physically consistent schedules while coordinating renewable generation, stationary storage, mobile EV resources, and grid exchanges across parameter-heterogeneous microgrids.

5.3.4. EV mobility dynamics and dispatchable availability

To provide direct evidence of how EV mobility affects MMG scheduling, this subsection examines the time-varying mobility states, charging-access availability, queuing evolution, and dispatchable power capacity of the EV population. The analysis is conducted on the same representative test day used in the scheduling and SOC studies.

Fig. 6(a) reveals a pronounced diurnal transition among the traveling, queuing, and connected EV states. During the early morning, most EVs remain connected to the residential MMGs, providing a relatively large pool of dispatchable flexibility. As the morning departure period begins, the connected population and the corresponding connection ratio decline rapidly, while the number of traveling EVs increases. After the departure peak, only a small fraction of the fleet remains available for charging or V2G operation, and EV flexibility becomes nearly unavailable during the daytime travel period. The reverse transition occurs after the evening return period, when the traveling population decreases and the connected population gradually recovers.

The microgrid-level results in Fig. 6(b) show that the connected-EV populations of MG1–MG3 follow similar temporal patterns but differ in magnitude because of their different EV population scales. MG1 consistently exhibits the largest connected population, followed by MG2 and MG3. Despite these scale differences, all three microgrids experience a sharp loss of EV availability during the morning departure window

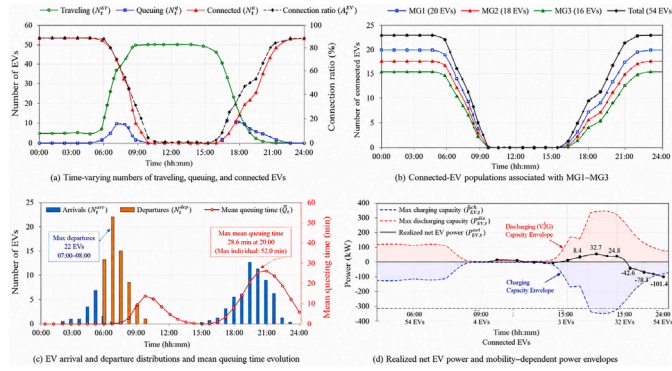


Fig. 6. EV mobility dynamics and their effects on dispatchable flexibility on the representative test day. (a) time-varying numbers of traveling, queuing, and physically connected EVs; (b) connected-EV populations associated with MG1–MG3; (c) EV arrival and departure distributions together with the evolution of mean queuing time; and (d) realized net EV power and the mobility-dependent charging and discharging capacity envelopes.

Table 9

Quantitative statistics of EV mobility and connection-dependent dispatchable availability.

Metric	MG1	MG2	MG3	System
Total number of EVs	20	18	16	54
Mean connected EVs	8.7	7.8	6.9	23.4
Peak connected EVs	20	18	16	54
Mean connection ratio (%)	43.5	43.3	43.1	43.3
Peak queuing population	5	4	4	13
Mean queuing time (min)	16.8	19.2	21.5	19.0
Maximum queuing time (min)	43.5	47.0	52.0	52.0
Unavailable-state EV exchange (kWh)	$<10^{-6}$	$<10^{-6}$	$<10^{-6}$	$<10^{-6}$

and a progressive recovery after the evening arrivals. This synchronized mobility pattern creates a system-wide contraction and subsequent restoration of flexible EV resources, while the unequal fleet sizes lead to different local charging and V2G potentials.

Fig. 6(c) further clarifies the temporal origin of these availability changes. The departure flow is concentrated in the morning, with the largest hourly departure volume reaching 22 EVs between 07:00 and 08:00. In contrast, arrivals are distributed over the late-afternoon and evening periods, producing a broader return profile. Because EV arrivals do not immediately imply physical charging access, the evening return process is accompanied by a temporary increase in queuing time. The mean queuing time reaches its maximum value of 28.6 min at approximately 20:00, while the maximum individual waiting time reaches 52.0 min. This delay between arrival and connection demonstrates that mobility availability and charging accessibility are distinct quantities: an EV may have reached a microgrid but may still be unavailable for energy exchange.

The mobility-dependent consequences for power scheduling are illustrated in Fig. 6(d). The charging and discharging envelopes contract substantially during the daytime period as the connected-EV population declines, and expand again after the evening return process. The realized net EV power remains within these time-varying envelopes, confirming that the learned policy respects the physical availability imposed by EV connection states. During the evening high-load and high-price period, the connected EV fleet provides net V2G support of 8.4, 32.7, and 24.8 kW at representative intervals. As the tariff decreases and more EVs regain charging access, the direction of power exchange reverses, with net charging powers of 42.6, 78.3, and 101.4 kW during the later evening hours.

Table 9 provides further quantitative evidence of the mobility-dependent EV availability. The mean connection ratios of MG1, MG2,

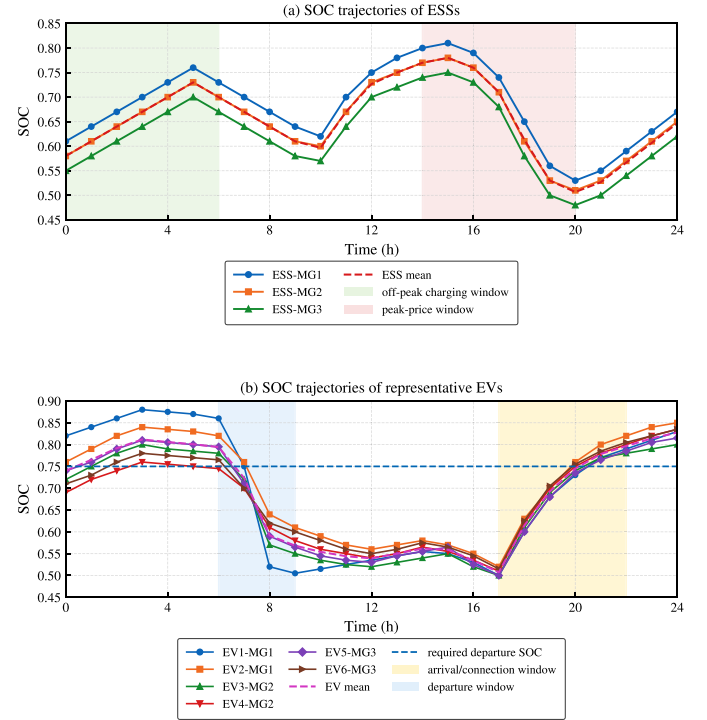


Fig. 7. SOC trajectories of ESSs and representative EVs on a test day. (a) SOC trajectories of ESSs in different microgrids, together with the ESS SOC range, mean trajectory, off-peak charging window, and peak-price window. (b) SOC trajectories of representative EVs, together with the EV SOC range, mean trajectory, required departure SOC, arrival/connection window, and departure window.

and MG3 are 43.5%, 43.3%, and 43.1%, respectively, indicating that less than half of the nominal EV population is available for energy scheduling on average. MG3 exhibits the longest mean and maximum queuing times because its EV population is relatively large with respect to its charging-access capacity. Nevertheless, the accumulated charging/discharging energy assigned to traveling or queuing EVs is below 10^{-6} kWh for all three microgrids, confirming that the connection constraint is respected within numerical tolerance.

The above results establish a direct causal chain from EV mobility to MMG scheduling. Traveling and queuing determine the physically connected population; the connected population determines the feasible charging and V2G capacity; and the resulting capacity envelope constrains the power that can be coordinated by the microgrid agents. The proposed method therefore adapts its scheduling decisions to the time-varying spatio-temporal availability of EVs rather than treating them as continuously accessible stationary storage resources.

5.3.5. SOC behaviors of ESSs and EVs

Fig. 7 shows the SOC trajectories of ESSs and representative EVs on a test day. In Fig. 7(a), the ESSs are charged during the off-peak charging window of 0:00–6:00, and their SOC levels increase from 0.55–0.61 to 0.70–0.76. During the peak-price window of 14:00–20:00, the ESSs are discharged to support local demand and reduce grid imports, resulting in a clear SOC decrease from 0.75–0.81 to 0.48–0.53. The three ESSs exhibit similar temporal patterns while maintaining different SOC levels, indicating coordinated storage dispatch across heterogeneous microgrids.

Fig. 7(b) presents the SOC trajectories of representative EVs. The EV departure window is set to 6:00–9:00, during which EVs leave the microgrids for travel and their SOC levels decrease accordingly. The arrival/connection window is set to 17:00–22:00, after which the EVs are reconnected and charged to recover their SOC levels. The SOC levels of representative EVs

Table 10
Ablation study results.

Variant	J	J_{MG}	J_{EV}	V_{bal}	η_{dep}	$\bar{R}^{risk}$
Ours	14,392.7	10,610.5	3782.2	1.2	99.0	0.023
w/o EV attention	14,864.9	10,895.7	3969.2	2.3	98.1	0.034
w/o feasibility mapping	15,106.3	10,974.8	4131.5	13.4	96.2	0.041
w/o EV-aware regularization	14,610.5	10,572.1	4038.4	1.9	96.8	0.052
w/o twin critics	14,790.2	10,828.6	3961.6	3.7	97.4	0.039

Table 11
Sensitivity analysis with respect to EV-side objective weight.

ω_{EV}	J_{MG} (\$)	J_{EV} (\$)	η_{dep} (%)	$\bar{R}^{risk}$	EV V2G (kWh)
0.2	10,380.4	4455.7	95.8	0.066	821.3
0.5	10,492.8	4102.5	97.3	0.046	703.8
1.0	10,610.5	3782.2	99.0	0.023	574.6
1.5	10,830.9	3570.8	99.4	0.017	486.2
2.0	11,152.6	3468.1	99.8	0.012	391.7

reach 0.80–0.85 by the end of the day, exceeding the required departure SOC of 0.75. These results show that the proposed policy can exploit EV flexibility for energy management while satisfying mobility requirements.

Overall, the SOC trajectories reveal the complementary roles of stationary and mobile storage resources. ESSs primarily provide intraday energy shifting between off-peak and peak-price periods, whereas EVs participate in charging and V2G operation subject to their arrival, connection, and departure constraints. The results confirm that the proposed MA-DRL framework can coordinate stationary and mobile storage resources across parameter-heterogeneous microgrids while maintaining operational feasibility and EV service reliability.

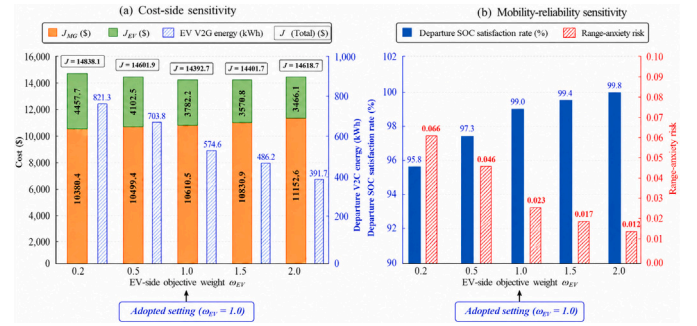
5.3.6. Ablation study

Table 10 reports the ablation results of the proposed method. Removing the EV-aware attention module increases the total objective value from 14,392.7 to 14,864.9, corresponding to a 3.28% degradation. Meanwhile, J_{EV} increases by 4.94%, and $\bar{R}^{risk}$ rises from 0.023 to 0.034, indicating that the attention module is effective in capturing heterogeneous EV mobility-energy states.

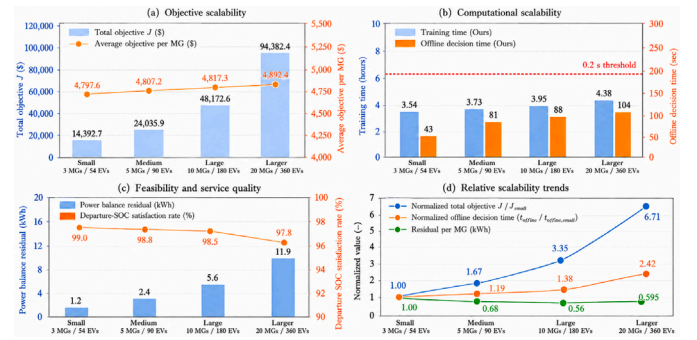
The feasibility-aware mapping layer has the most significant impact on constraint satisfaction. Without this module, V_{bal} increases from 1.2 to 13.4 kWh, i.e., more than an elevenfold increase, while η_{dep} decreases from 99.0% to 96.2%. This confirms that embedding physical feasibility into the action-generation process is essential for maintaining reliable microgrid operation. Removing the EV-aware regularization slightly reduces J_{MG} from 10,610.5 to 10,572.1, but increases J_{EV} by 6.77% and raises $\bar{R}^{risk}$ from 0.023 to 0.052. This result shows that the regularization term effectively prevents excessive V2G discharging at the expense of EV mobility reliability. In addition, removing the twin critics increases J by 2.76% and V_{bal} from 1.2 to 3.7 kWh, demonstrating the benefit of conservative value estimation in the hybrid multi-agent action space.

5.3.7. Sensitivity to EV-side weight

Table 11 and Fig. 8 present the sensitivity of the proposed method to the EV-side objective weight ω_{EV} . As ω_{EV} increases from 0.2 to 2.0, the EV-side cost decreases from \$4455.7 to \$3468.1, corresponding to a 22.16% reduction, while the microgrid-side cost increases from \$10,380.4 to \$11,152.6, corresponding to a 7.44% increase. This trend indicates a clear trade-off between microgrid operating economy and EV mobility reliability. The EV-side reliability indicators improve consistently with a larger ω_{EV} . Specifically, the departure-SOC satisfaction rate increases from 95.8% to 99.8%, while the average range-anxiety risk decreases from 0.066 to 0.012. Meanwhile, the EV V2G energy decreases from 821.3 kWh to 391.7 kWh, indicating that a larger EV-side

**Fig. 8.** Sensitivity analysis with respect to the EV-side objective weight. (a) cost-side sensitivity in terms of J_{MG} , J_{EV} , total objective value J , and EV V2G energy. (b) mobility-reliability sensitivity in terms of departure-SOC satisfaction rate and range-anxiety risk.**Table 12**
Scalability analysis under different system sizes.

Scale	MGs	EVs	J	V_{bal}	η_{dep}	Train (h)	T_{dec} (s)
Small	3	54	14,392.7	1.2	99.0	3.54	0.043
Medium	5	90	24,035.9	2.4	98.8	3.73	0.081
Large	10	180	48,172.6	5.6	98.5	3.95	0.088
Larger	20	360	96,588.4	11.9	97.8	4.38	0.104

**Fig. 9.** Scalability analysis of the proposed method under different system sizes. (a) objective scalability in terms of the total objective value and the average objective value per microgrid. (b) computational scalability in terms of training time and online decision time. (c) feasibility and service quality under increasing system size. (d) relative scalability trends of the normalized objective value, normalized online decision time, and power-balance residual per microgrid.

weight suppresses aggressive EV discharging and reserves more battery energy for subsequent trips.

Among all tested settings, $\omega_{EV} = 1.0$ yields the lowest total objective value of \$14,392.7. Compared with $\omega_{EV} = 0.2$, this setting reduces the total objective value by 3.00%, decreases the EV-side cost by 15.12%, and lowers the range-anxiety risk by 65.15%. Further increasing ω_{EV} from 1.0 to 2.0 improves the departure-SOC satisfaction rate by only 0.8 percentage points but increases the total objective value by 1.57%. Therefore, $\omega_{EV} = 1.0$ provides a balanced compromise between economic operation and EV-side service reliability and is adopted as the default setting in the following experiments.

5.3.8. Scalability and robustness

The scalability results in Table 12 and Fig. 9 demonstrate that the proposed method can be effectively extended to larger MMG systems. When the system size increases from 3 microgrids with 54 EVs to 20 microgrids with 360 EVs, the total objective value increases from \$14,392.7 to \$96,588.4, corresponding to a 6.71-fold increase. This growth is consistent with the 6.67-fold expansion in system scale.

Table 13
Robustness analysis under renewable and EV mobility uncertainty.

RES Unc.	EV Unc.	J (\$)	V_{bal} (kWh)	η_{dep} (%)	$\bar{R}^{risk}$
0%	0%	14,392.7	1.2	99.0	0.023
5%	5%	14,608.5	1.6	98.7	0.026
10%	10%	14,951.2	2.5	98.1	0.032
20%	20%	15,582.6	4.7	96.9	0.044

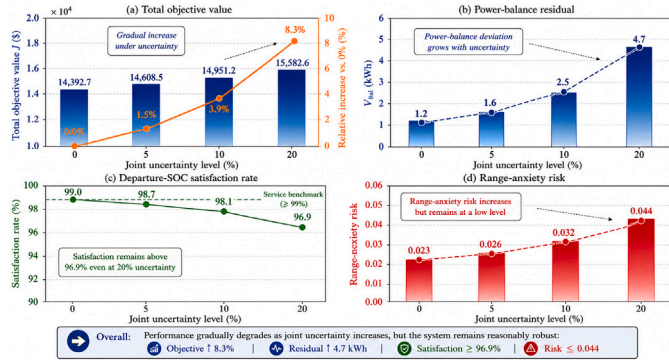


Fig. 10. Robustness analysis under renewable-generation and EV-mobility uncertainties. (a) total objective value and its relative increase with respect to the nominal case. (b) power-balance residuals under different joint uncertainty levels. (c) departure-SOC satisfaction rate of EVs. (d) range-anxiety risk.

More importantly, the average objective value per microgrid remains nearly stable, increasing only from \$4797.6 to \$4829.4, i.e., by 0.66%. This indicates that the proposed method maintains stable per-microgrid scheduling performance under system expansion.

The computational results further confirm the scalability of decentralized execution. The online decision time increases from 0.043 s to 0.104 s as the number of microgrids increases from 3 to 20, corresponding to a 2.42-fold increase, but it remains below the 0.2 s real-time scheduling threshold. The training time increases from 3.54 h to 4.38 h, with only a 23.73% increase, showing that the computational burden grows moderately relative to the system scale. In terms of feasibility, the total power-balance residual increases from 1.2 kWh to 11.9 kWh; however, the residual per microgrid remains limited, ranging from 0.40 kWh to 0.595 kWh. Meanwhile, the departure-SOC satisfaction rate decreases only from 99.0% to 97.8%, remaining at or above 97.8% in all tested cases. These results verify that the proposed framework achieves favorable scalability in terms of economic performance, computational efficiency, constraint satisfaction, and EV service reliability.

To evaluate robustness, renewable generation and EV mobility profiles are perturbed simultaneously. Table 13 and Fig. 10 report the performance of the proposed method under different joint uncertainty levels. As the uncertainty level increases from 0% to 20%, the total objective value increases from \$14,392.7 to \$15,582.6, corresponding to an 8.3% increase relative to the nominal case. This moderate degradation indicates that the proposed method can maintain stable economic performance under renewable-generation and EV-mobility uncertainty.

The feasibility and EV-side reliability indicators also remain within acceptable ranges. The power-balance residual increases from 1.2 kWh to 4.7 kWh as the uncertainty level increases to 20%, showing that uncertainty introduces additional power mismatch. Nevertheless, the residual remains limited compared with the system operating scale. Meanwhile, the departure-SOC satisfaction rate decreases only from 99.0% to 96.9%, i.e., by 2.1 percentage points, while the range-anxiety risk increases from 0.023 to 0.044. Even under 20% joint uncertainty, the satisfaction rate remains above 96% and the range-anxiety risk remains below 0.05. These results confirm that the proposed method preserves economic

Table 14
Low-carbon performance comparison.

Method	$E_{CO_2}^{total}$ (kg)	RES Util. (%)	Import (kWh)	Export (kWh)
MPC	856.4	91.8	4876.5	612.8
MADDPG	925.7	88.5	5250.3	498.6
Independent DRL	890.1	89.7	5064.8	535.4
Centralized MASAC	862.5	91.2	4902.6	584.1
MAAC-PER	839.6	92.6	4768.3	640.5
Ours	796.5	95.4	4526.7	708.9

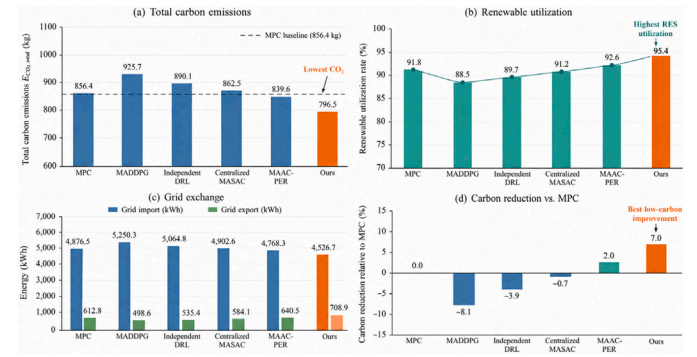


Fig. 11. Low-carbon performance comparison of different methods. (a) total carbon emissions of different methods. (b) renewable-energy utilization rate. (c) grid Import and grid Export energy. (d) carbon-emission reduction relative to MPC.

performance, operational feasibility, and EV mobility reliability under simultaneous renewable-generation and mobility perturbations.

5.3.9. Low-carbon performance

Table 14 and Fig. 11 compare the low-carbon performance of different methods. The proposed method achieves the lowest total carbon emissions of 796.5 kg and the highest renewable-energy utilization rate of 95.4%. Compared with MAAC-PER, which is the strongest learning-based baseline in terms of carbon performance, the proposed method reduces $E_{CO_2}^{total}$ from 839.6 kg to 796.5 kg, corresponding to a 5.13% reduction. Compared with MPC and MADDPG, the carbon emissions are reduced by 7.00% and 13.96%, respectively.

The improvement is also reflected in renewable-energy utilization and grid exchange. The proposed method improves the renewable utilization rate by 2.8 percentage points compared with MAAC-PER and by 3.6 percentage points compared with MPC. Meanwhile, the grid import is reduced to 4526.7 kWh, which is 5.07% lower than that of MAAC-PER and 7.17% lower than that of MPC. In addition, the grid export increases to 708.9 kWh, exceeding MAAC-PER and MPC by 10.68% and 15.68%, respectively. These results indicate that the proposed method can more effectively coordinate renewable generation, ESSs, EV flexibility, demand response, and grid transactions, thereby enhancing renewable-energy absorption and reducing carbon emissions in MMG energy management.

Overall, the case studies provide comprehensive evidence for the effectiveness of the proposed HEC-MADRL framework. The comparative results demonstrate its advantages in economic operation, low-carbon performance, constraint satisfaction, EV mobility reliability, and online decision efficiency. The ablation study further verifies the necessity of the EV-aware representation, hybrid actor design, feasibility-aware action mapping, EV-side regularization, and centralized twin-critic learning. In addition, the sensitivity, scalability, and uncertainty analyses confirm that the proposed framework can maintain stable performance under different objective preferences, system scales, and renewable-generation and EV-mobility uncertainties. These results indicate that the proposed method provides a scalable, reliable, and physically consistent

solution for real-time MMG energy management with mobile EV resources.

6. Conclusion

This paper proposed an HEC-MADRL framework for coordinated microgrid-EV mobility-energy management. A two-layer formulation was developed to link MMG operation with EV mobility behavior through aggregated EV charging/discharging powers, and the resulting scheduling problem was reformulated as a cooperative multi-agent MDP. Based on this formulation, a CTDE-based learning algorithm was designed by integrating EV-aware aggregation, hybrid decentralized policy learning, centralized twin-critic value estimation, feasibility-guided action processing, and EV-aware regularization, enabling scalable and physically consistent online scheduling.

Case studies on a residential MMG system demonstrated that HEC-MADRL outperforms representative baselines in economic operation, low-carbon performance, constraint satisfaction, EV mobility reliability, and online decision efficiency. Compared with the strongest MAACPER baseline, the proposed method reduced the total objective value by 4.43% and achieved an online decision time of 0.043 s. Future work will extend the framework to integrated multi-energy microgrids with heating, cooling, hydrogen, and flexible demand resources. More detailed EV user behavior, charging-service preferences, station congestion, and battery-aging models will also be incorporated to improve practical applicability. In addition, safe online adaptation and transfer learning will be investigated to enhance policy robustness under changing renewable generation, traffic conditions, market prices, and communication environments. Field-data-driven validation and hardware-in-the-loop testing will be further explored to support practical deployment in future low-carbon distribution networks.

CRedit authorship contribution statement

Qingshu Guan: Writing – original draft, Software, Methodology, Conceptualization. **Hui Cao:** Writing – review & editing, Supervision, Resources, Project administration, Methodology. **Xianjing Zhong:** Validation, Software, Investigation, Formal analysis. **Kai Qu:** Software, Methodology, Data curation. **Dapeng Yan:** Supervision, Project administration, Data curation. **Badong Chen:** Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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